

$$F = G \frac{m_1 m_2}{d^2}$$

$$\phi(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$i\hbar \frac{\partial}{\partial t} \psi = \hat{H} \psi$$

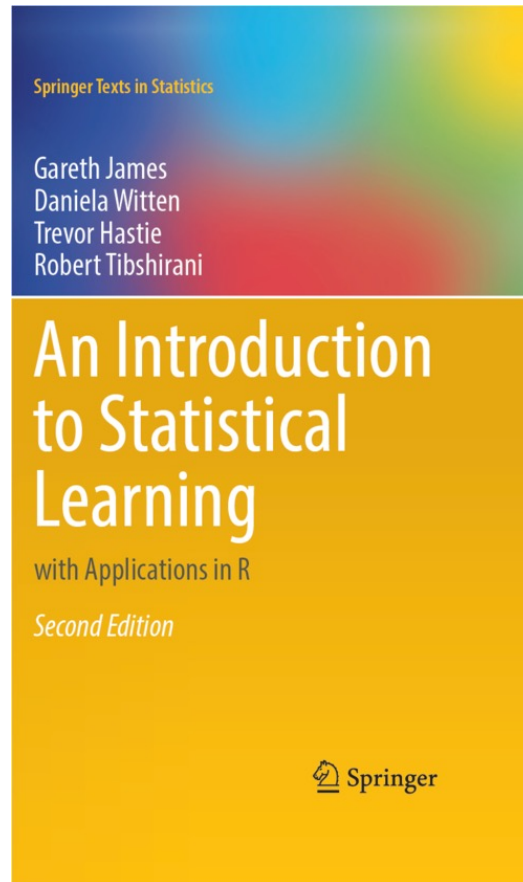
$$F - E + V = 2$$

Machine Learning

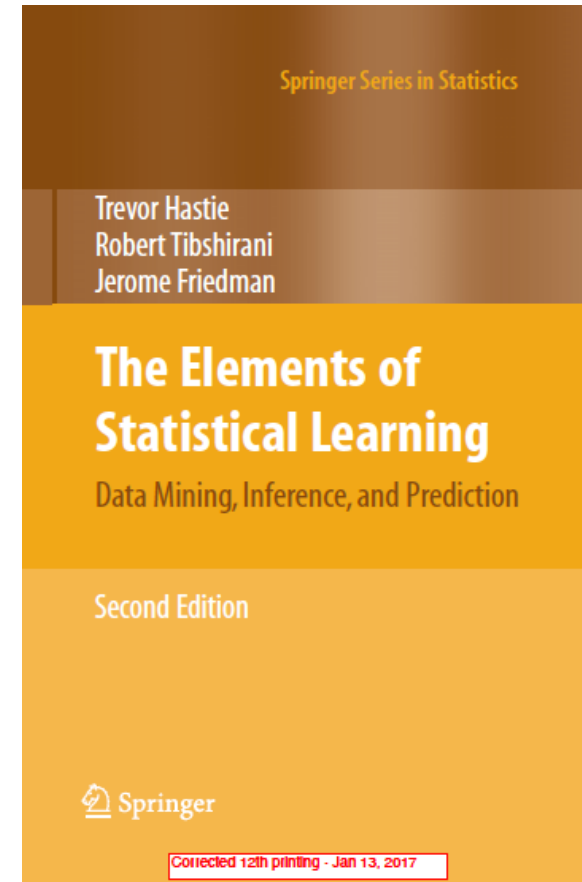
$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$$

$$\frac{df}{dt} = \lim_{h \rightarrow 0} \frac{f(t+h) - f(t)}{h}$$

Bibliografia



<https://www.statlearning.com/>



<https://hastie.su.domains/ElemStatLearn/>

The background is a dark, abstract composition. On the left, there is a glowing, translucent brain with a complex, textured surface. To the right of the brain, a dense, intricate network of white lines and dots, resembling a neural network or a complex geometric structure, extends across the upper and middle portions of the frame. The right side of the image fades into a bright white background where the title is located.

Visão geral sobre o Machine Learning

“Machine Learning é a área de estudo que dá habilidade a computadores de aprender algo que não foram explicitamente programados”

Arthur Samuel (1959)



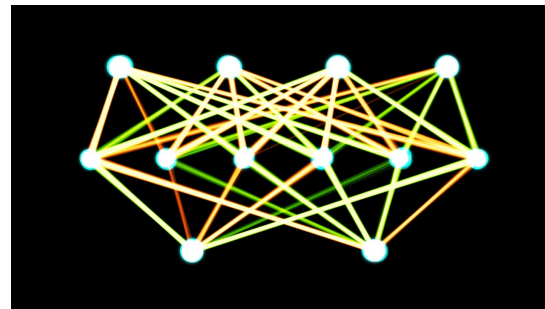
Evolução histórica do Machine Learning

Jogar xadrez



Arthur Samuel (1950-60)

Artificial Neural Network



Backpropagation (1973)

Deep learning landmark



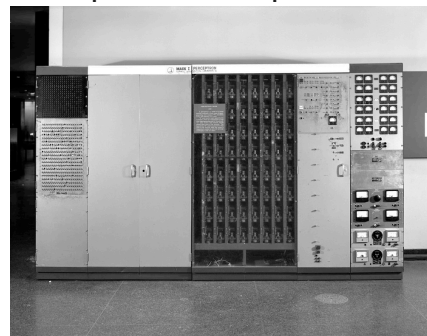
A. Krizhevsky, I. Sutskever, and G. E. Hinton (2012)

Teoria sobre o cérebro



Donald Hebb (1949)

Computador para ML



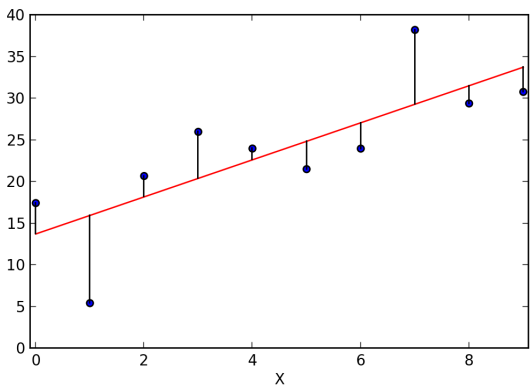
Mark 1 Perceptron (1958)

- Random Forests
 - Support Vector Machines
 - Deep Neural Networks
- (1980-1990)

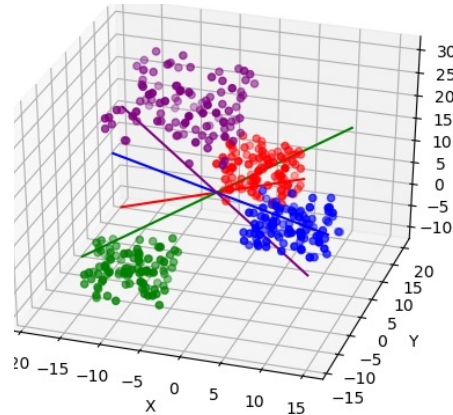


Evolução histórica do Machine (statistical) Learning

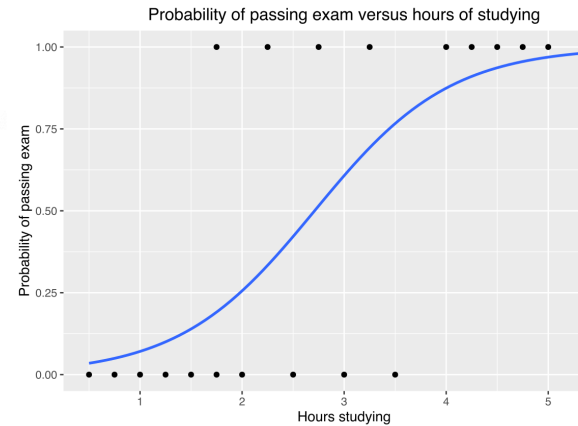
Linear Regression



Linear Discriminant Analysis



Logistic Regression



1800

1900

1930

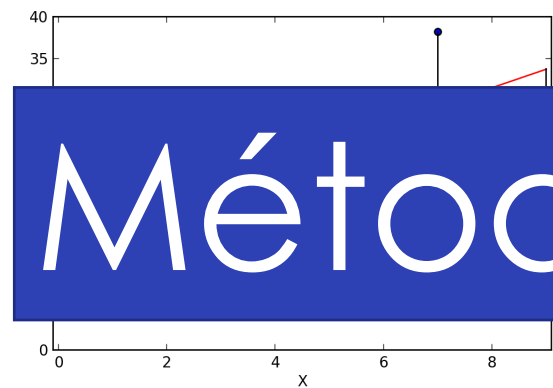
1960

1990

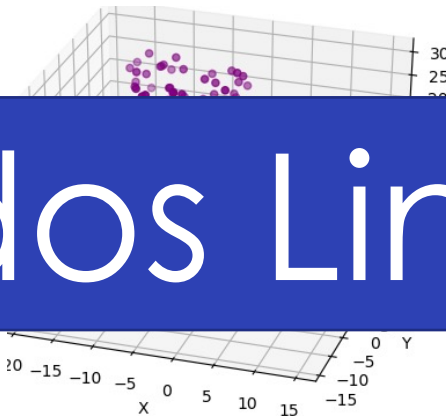
Presente

Evolução histórica do Machine (statistical) Learning

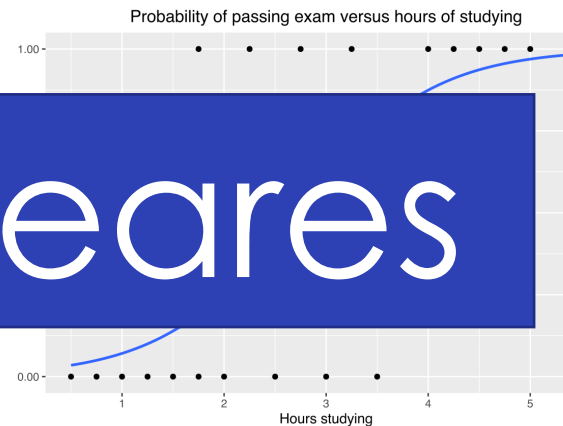
Linear Regression



Linear Discriminant Analysis



Logistic Regression



Métodos Lineares

1800

1900

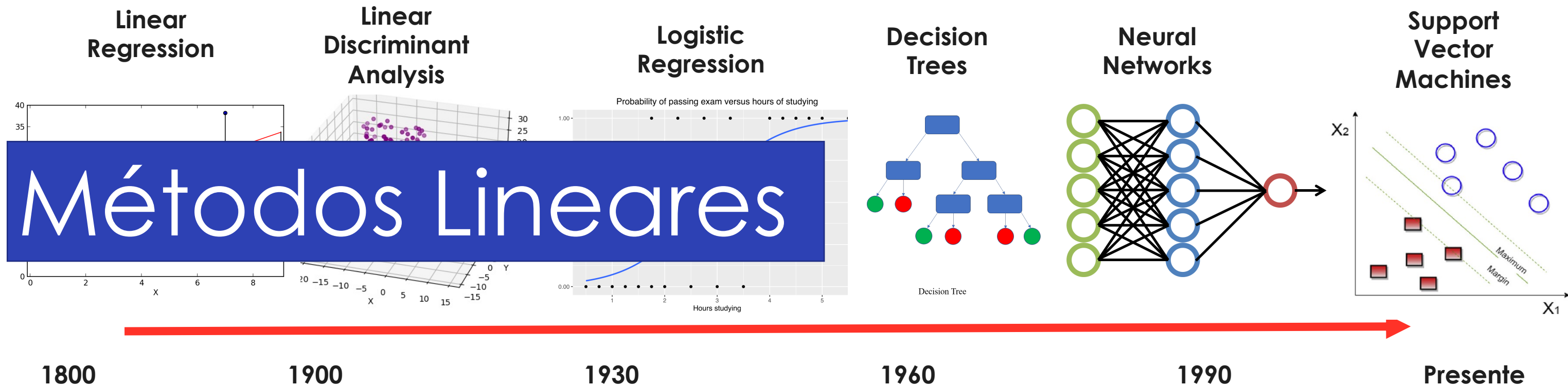
1930

1960

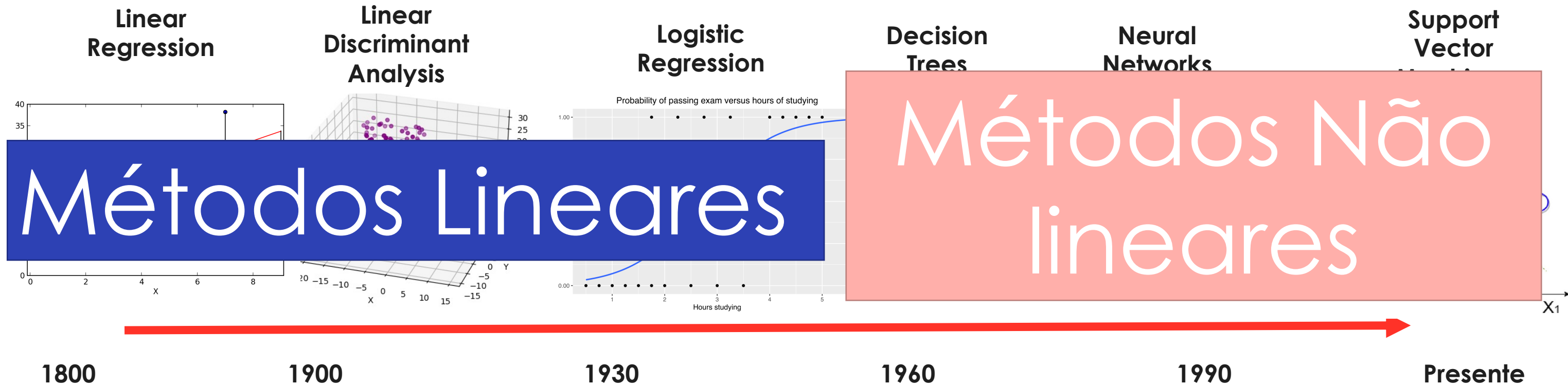
1990

Presente

Evolução histórica do Machine (statistical) Learning



Evolução histórica do Machine (statistical) Learning



Grande questão atual? Ética!!

Forbes

TECH

How Target Figured Out A Teen Girl Was Pregnant Before Her Father Did

Kashmir Hill Former Staff

Welcome to The Not-So Private Parts where technology & privacy collide

Feb 16, 2012, 11:02am EST

⌚ This article is more than 10 years old.



Target has got you in its aim

Every time you go shopping, you share intimate details about your consumption patterns with retailers. And many of those retailers are studying those details to figure out what you like, what you need, and which coupons are most likely to make you happy. [Target](#), for example, has figured out how to data-mine its way into your womb, to figure out whether you have a baby on the way long before you need to start buying diapers.

Charles Duhigg outlines in the [New York Times](#) how Target tries to hook

MICROSOFT / WEB / TL;DR

Twitter taught Microsoft's AI chatbot to be a racist asshole in less than a day



By JAMES VINCENT

Mar 24, 2016, 10:43 AM GMT | 0 Comments



It took less than 24 hours for Twitter to corrupt an innocent AI chatbot. Yesterday, Microsoft unveiled Tay — a Twitter bot that the company described as an experiment in "conversational understanding." The more you chat with Tay, said Microsoft, the smarter it gets, learning to engage people through "casual and playful conversation."

Unfortunately, the conversations didn't stay playful for long. Pretty soon after Tay launched, people starting tweeting the bot with all sorts of misogynistic, racist, and Donald Trumpist remarks. And

Forbes

DIVERSITY, EQUITY & INCLUSION

AI Bias Could Put Women's Lives At Risk - A Challenge For Regulators

Carmen Niethammer Former Contributor @

I am a private sector development expert and gender diversity leader.

Mar 2, 2020, 04:19am EST

When the European Commission released the long awaited white paper "[On Artificial Intelligence - A European approach to excellence and trust](#)" on February 19, much of the initial public reaction focused on potential AI regulation further challenging the EU's position in light of fierce technological competition from China and the United States.

Few discussed the European Commission's document mention of gender and ethical guidelines. Importantly, the white paper calls for "requirements to take reasonable measures aimed at ensuring that [the] use of AI systems does not lead to outcomes entailing prohibited discrimination." Why does it matter?

This is not simply about a theoretical approach to discrimination. It is largely also about saving (women's) lives - and ensuring that essential products and services meet the needs of both women and men. However, if artificial intelligence is based on "bad" data sourced predominantly from men and/or based on male profiles, terrible things can happen.

COLO
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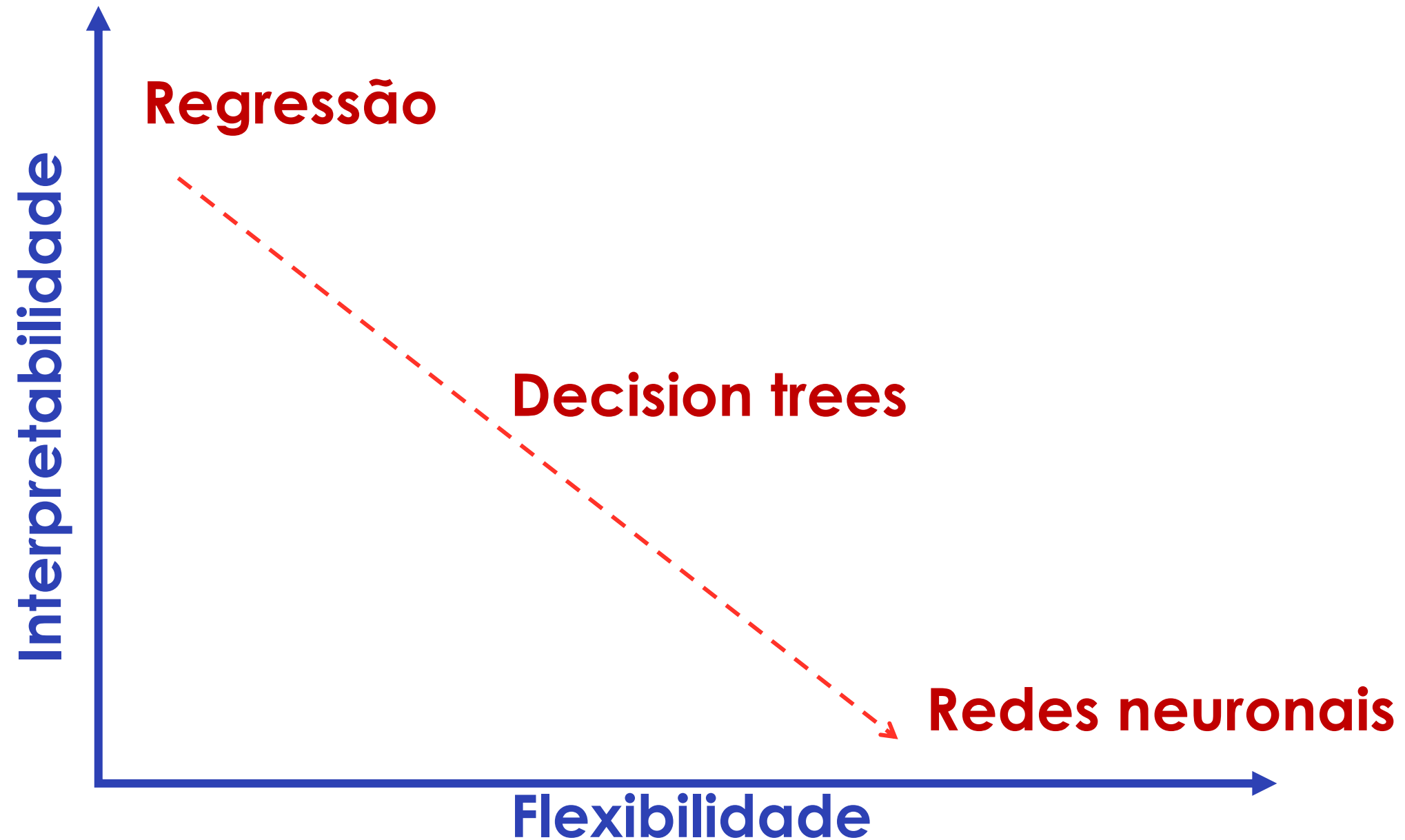
Compromissos na escolha de uma técnica de ML

- Precisão da previsão ou Interpretabilidade
(**accuracy**) (**inference**)
- Bom ajuste contra Sobreajuste ou Ajuste Insuficiente
(**good fit**) (**overfit**) (**underfit**)
- Paramétrico ou não-paramétrico

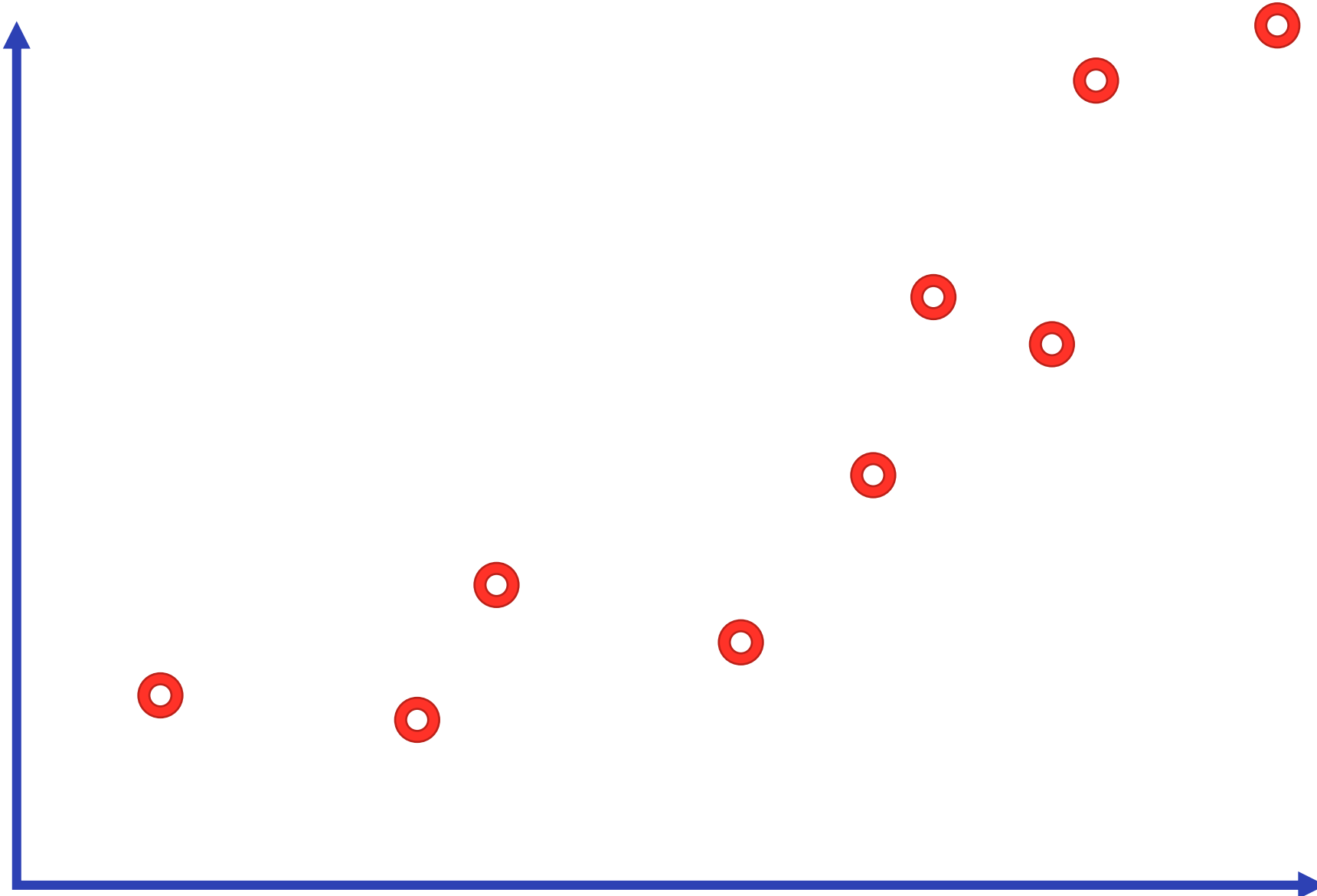
Compromissos na escolha de uma técnica de ML

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 - Paramétrico ou não-paramétrico
- Regressão ou classificação

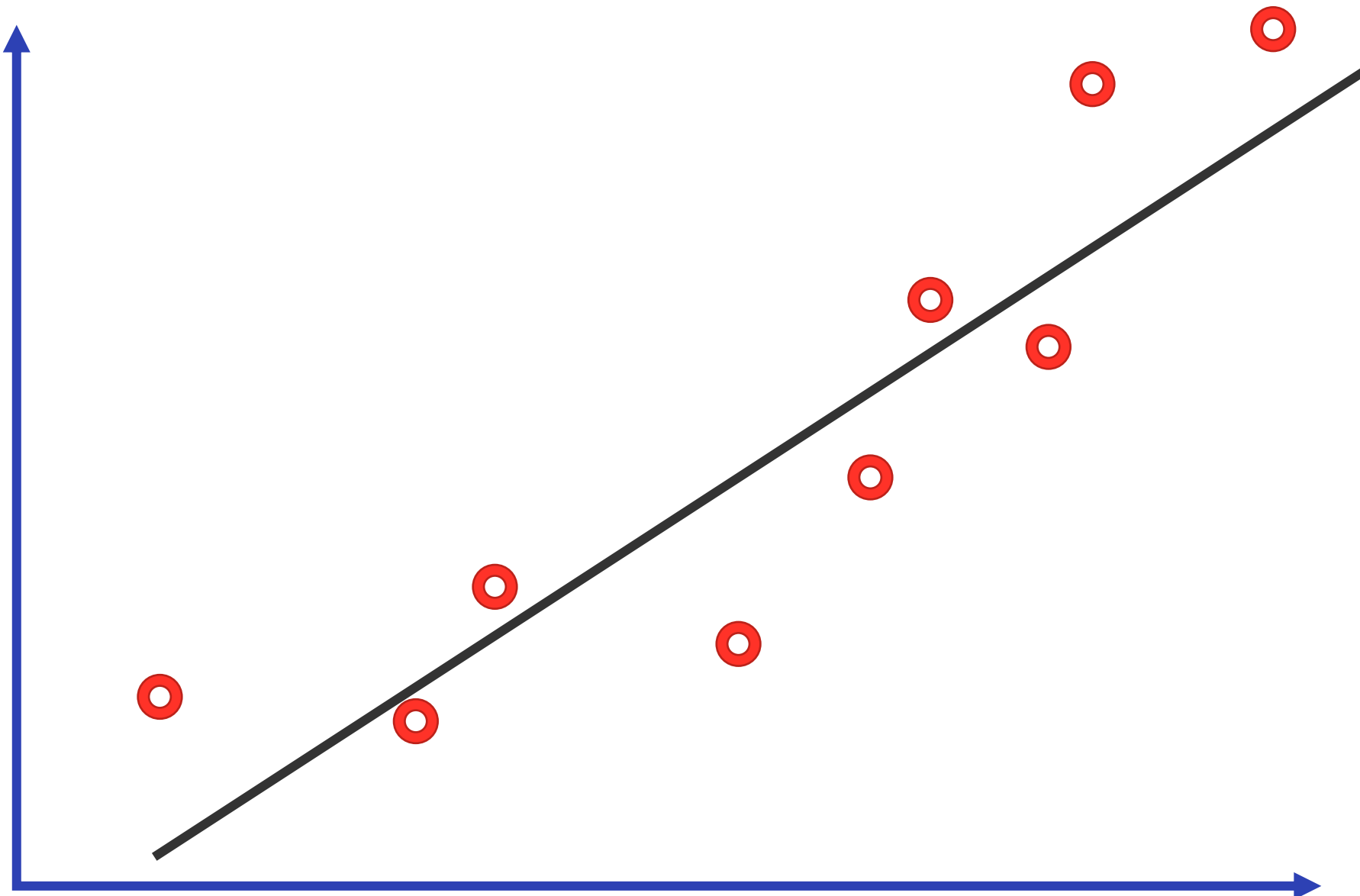
Flexibilidade ou Interpretabilidade



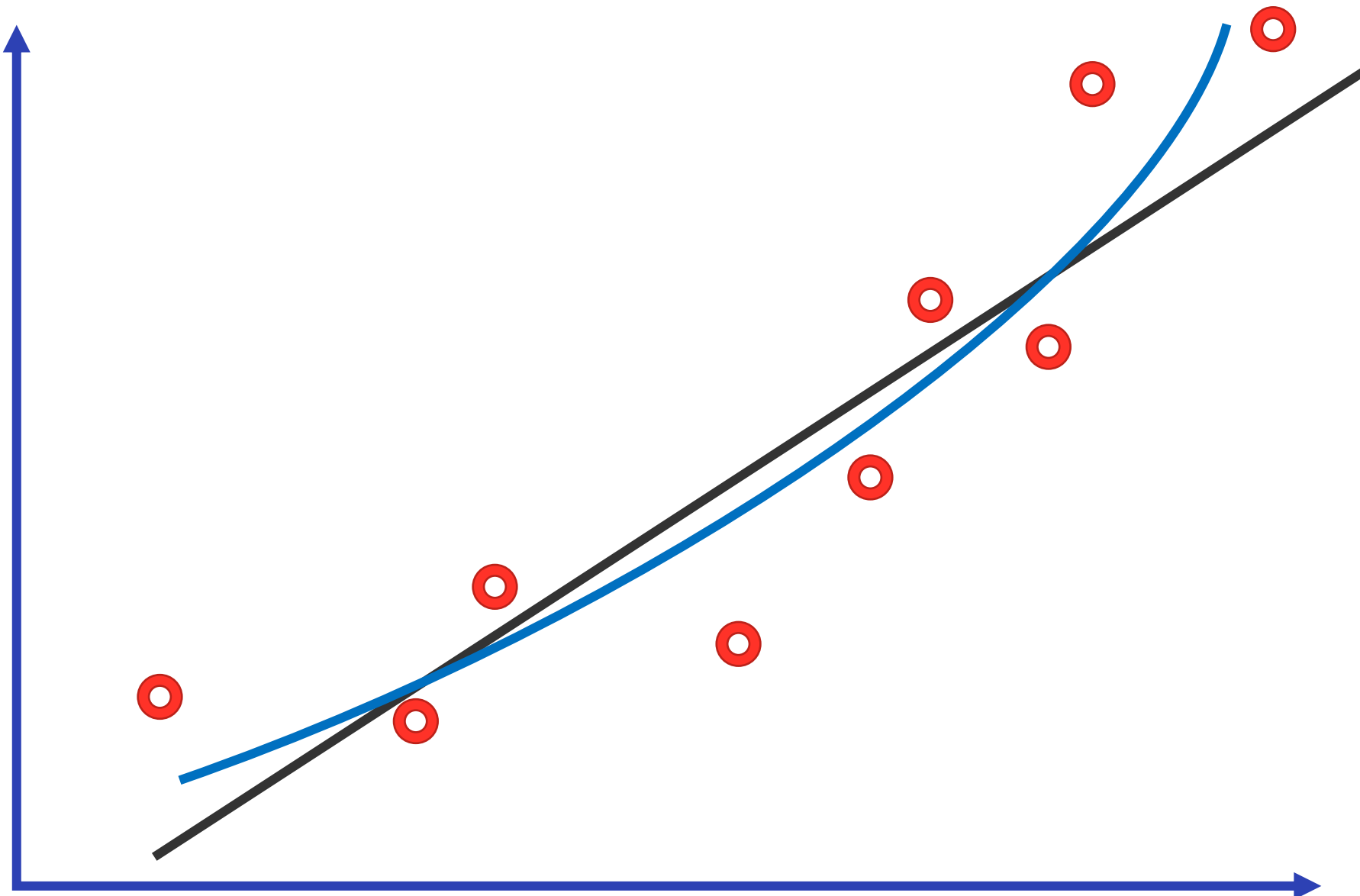
Como fazer um bom ajuste?



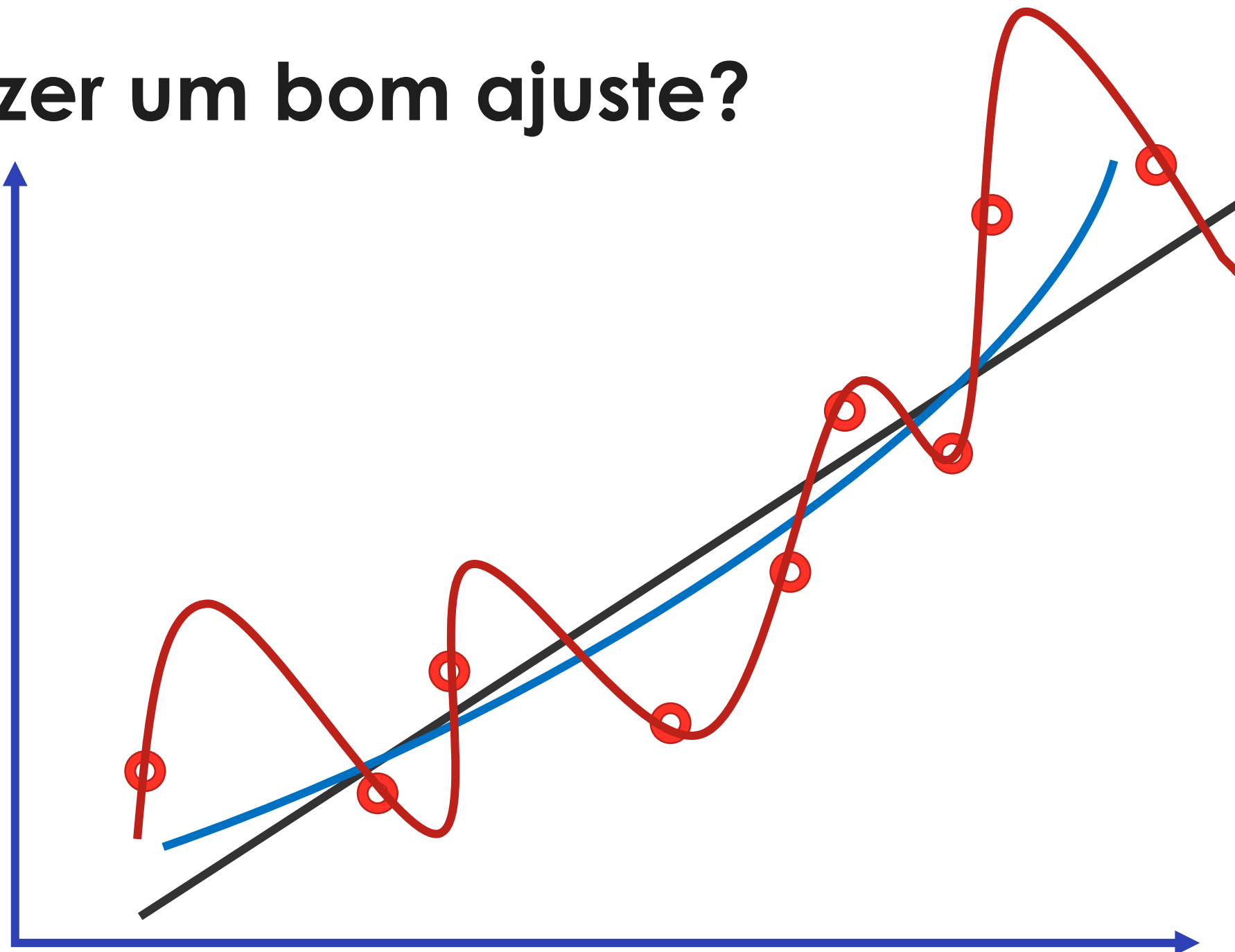
Como fazer um bom ajuste?



Como fazer um bom ajuste?



Como fazer um bom ajuste?

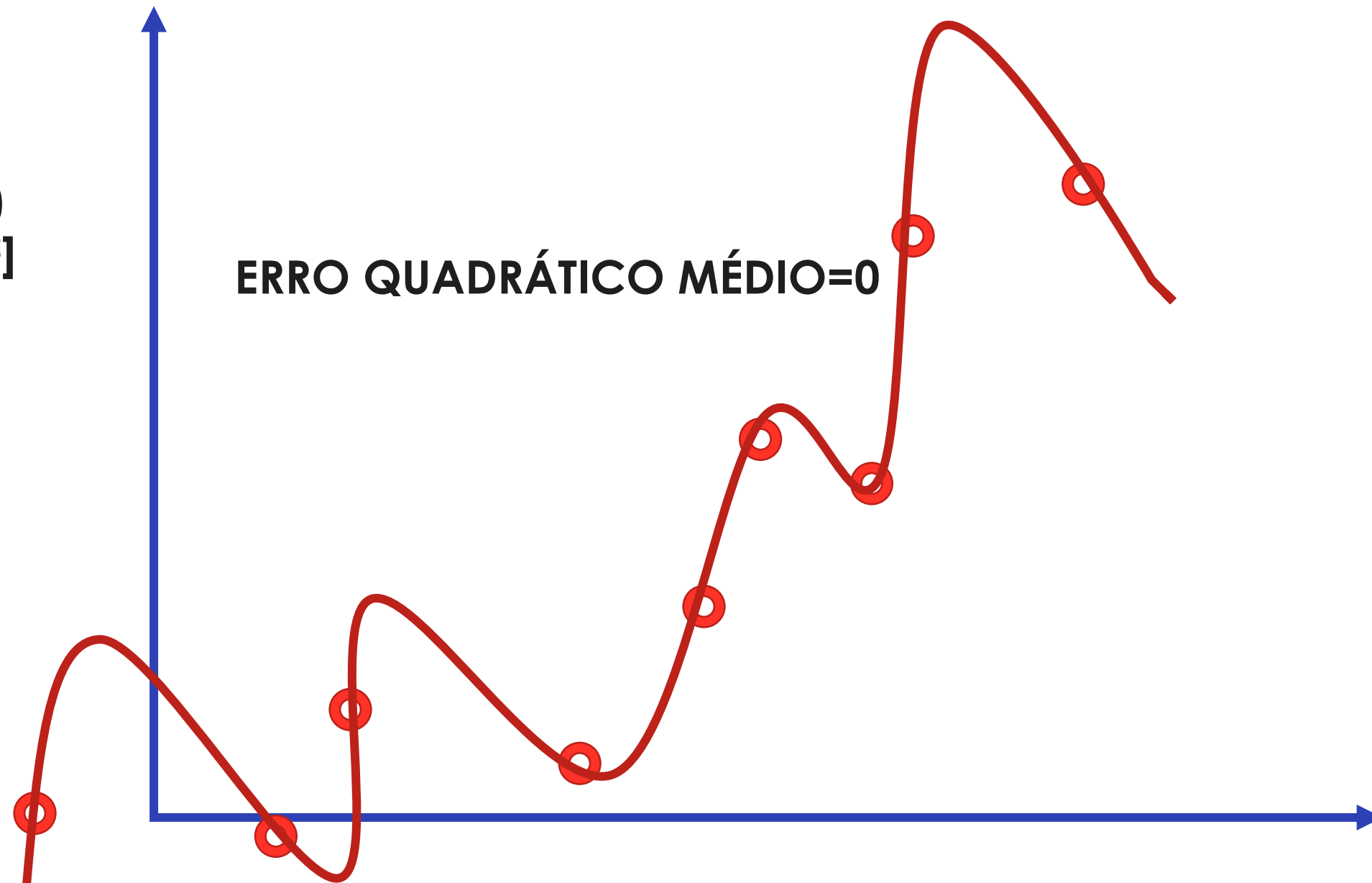


Como verificar a qualidade do ajuste?

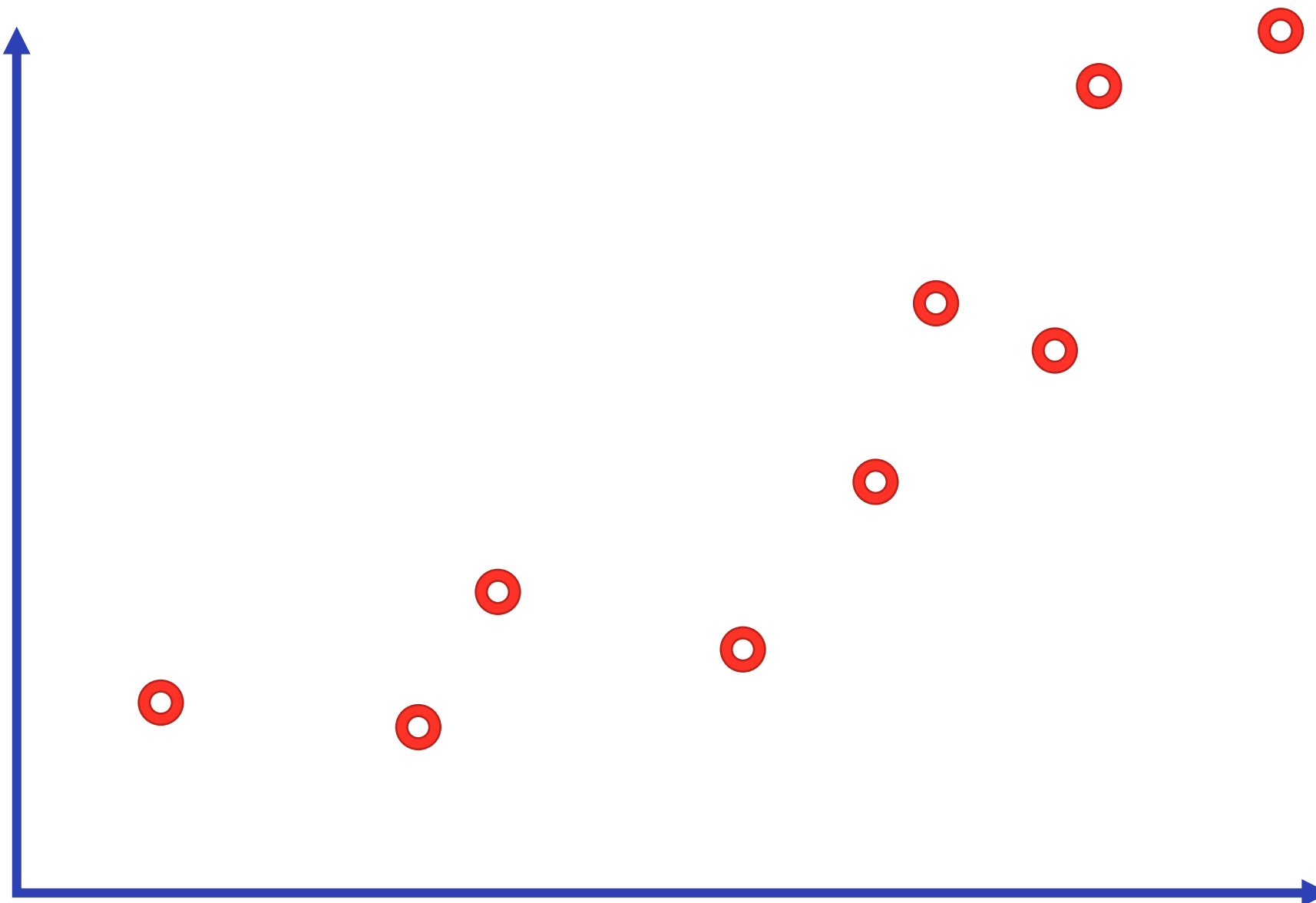
ERRO QUADRÁTICO MÉDIO (MSE)
= MÉDIA[(REAL(X) – MODELO(x))²]

Como verificar a qualidade do ajuste?

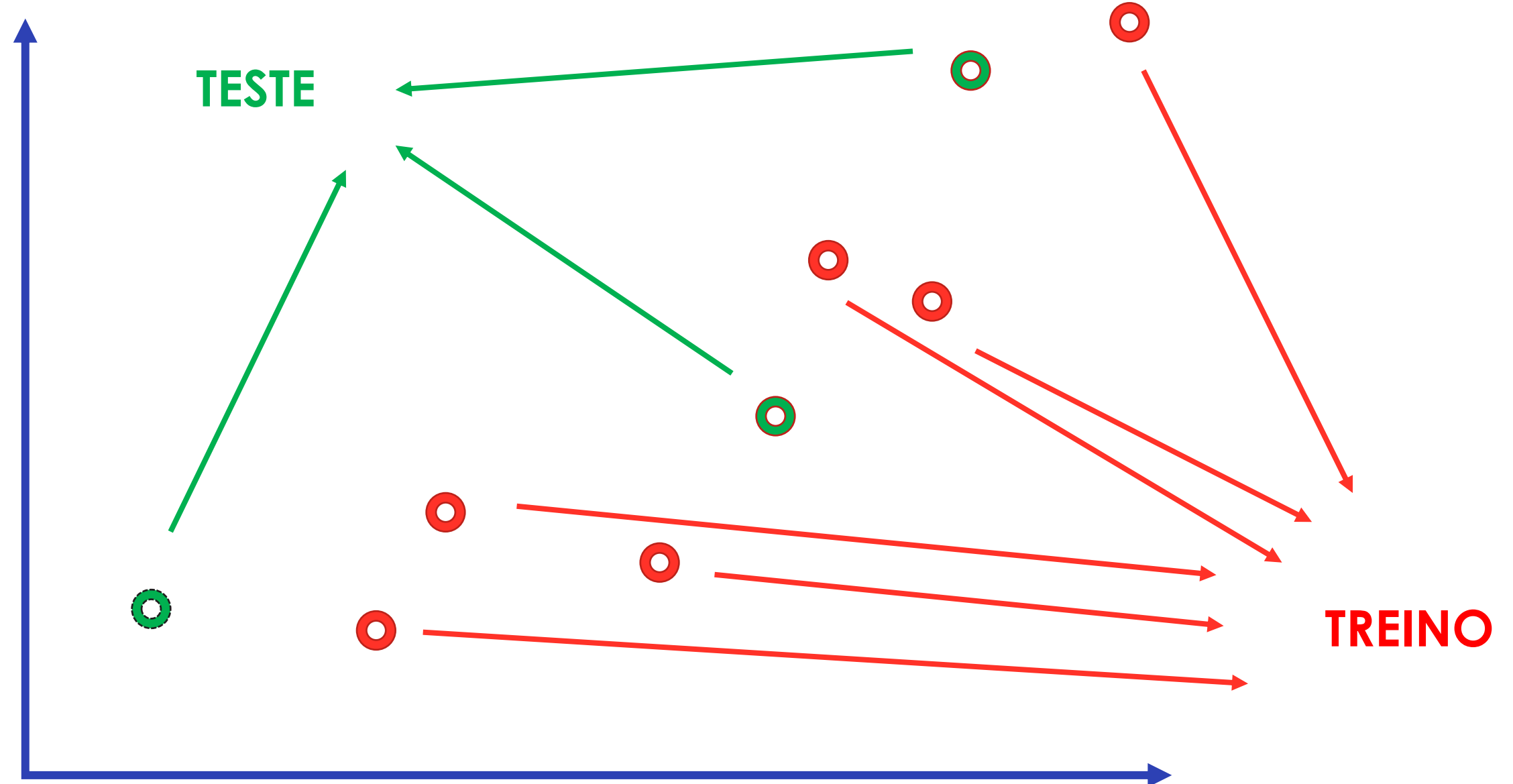
ERRO QUADRÁTICO MÉDIO (MSE)
= MÉDIA[(REAL(X) - MODELO(x))²]



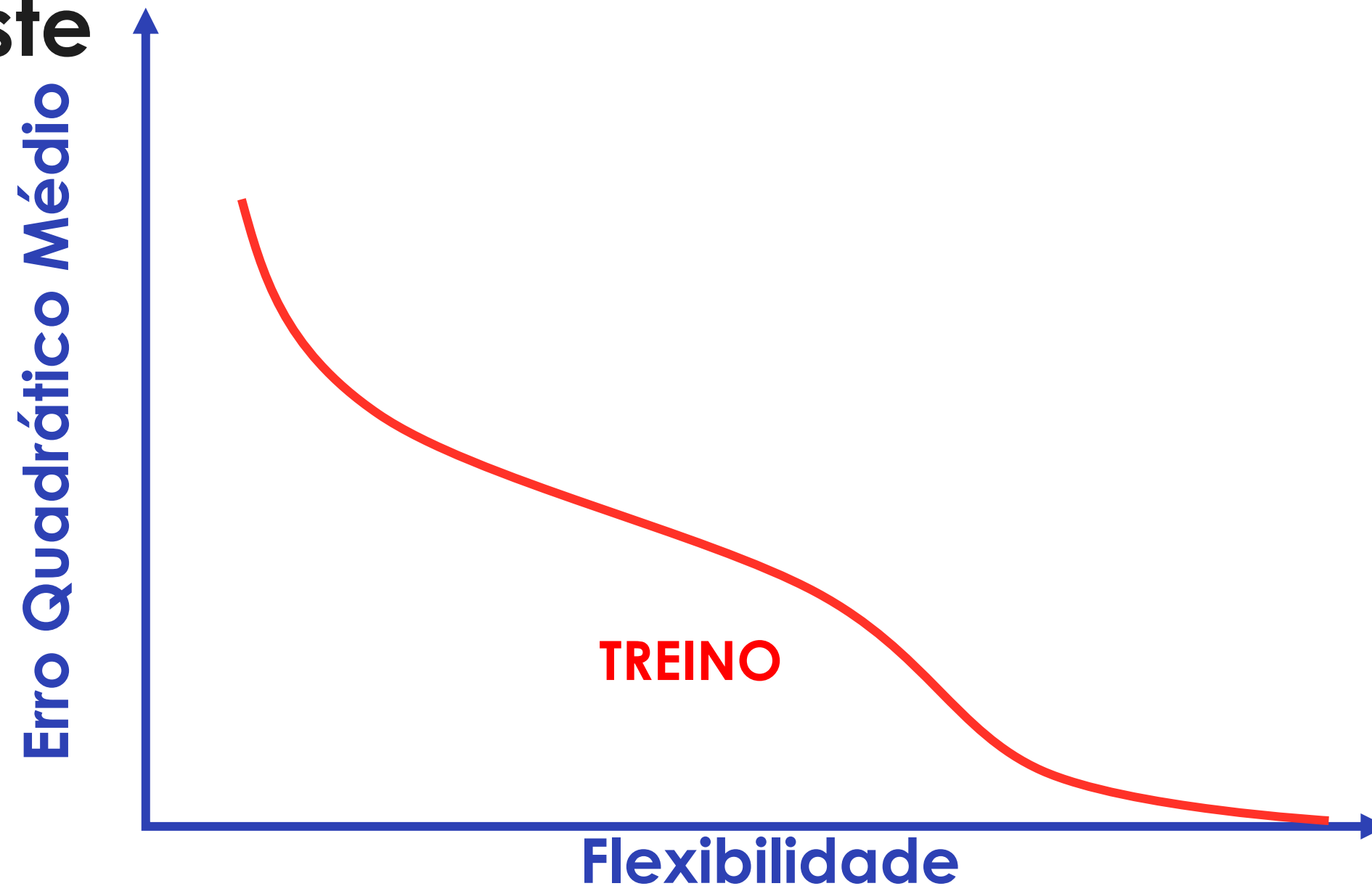
Separar os dados de treino dos dados de teste



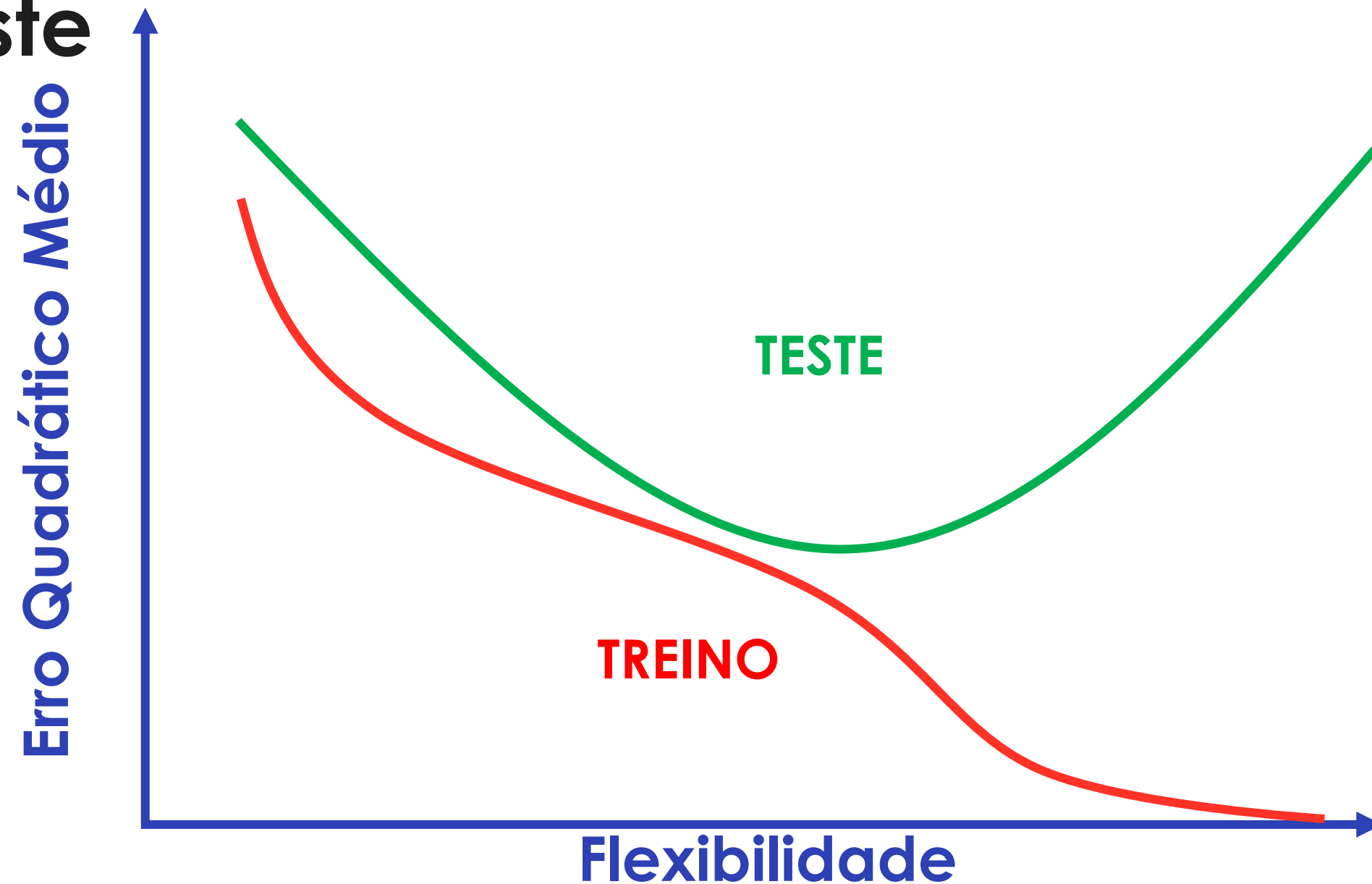
Separar os dados de treino dos dados de teste



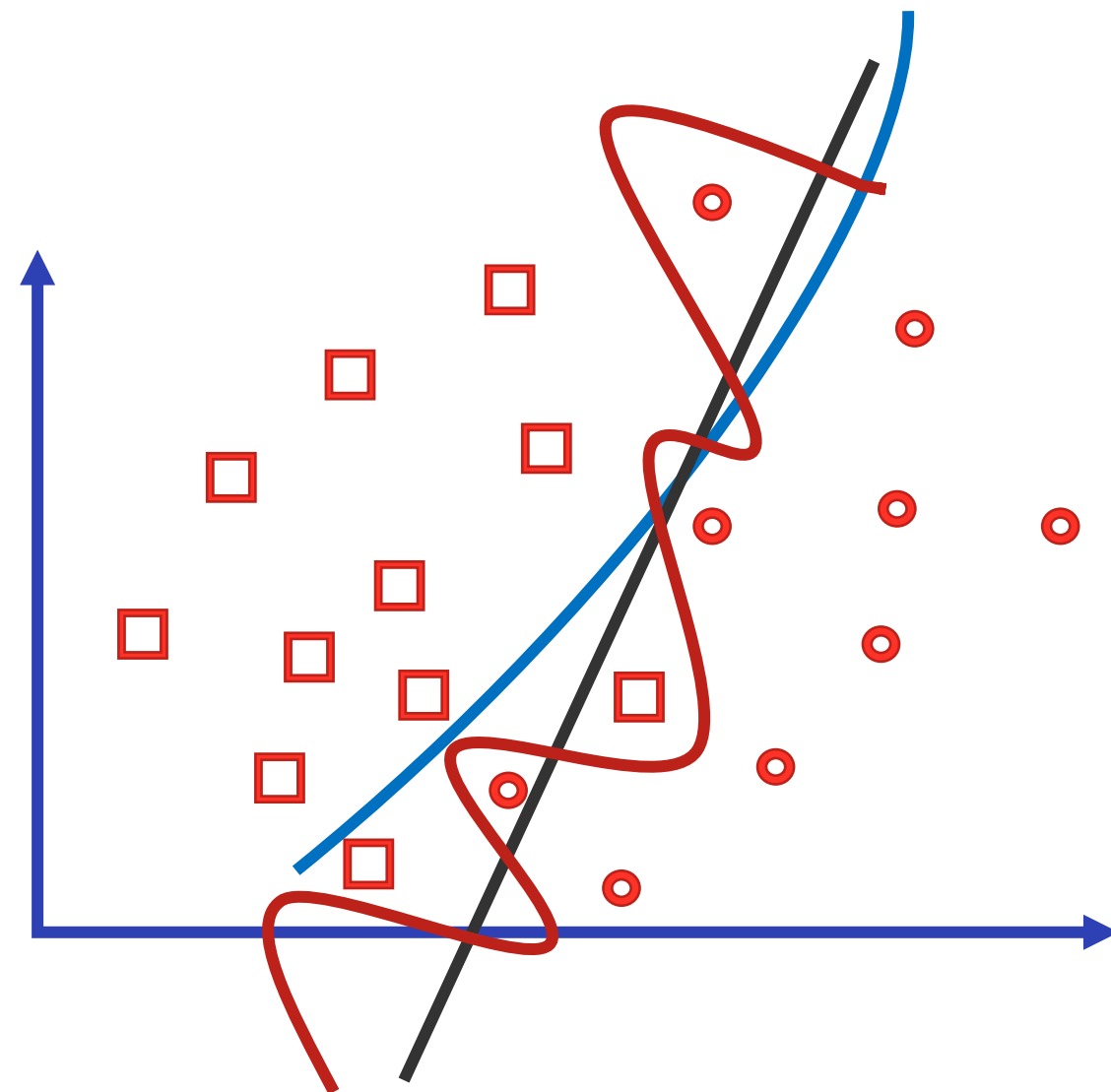
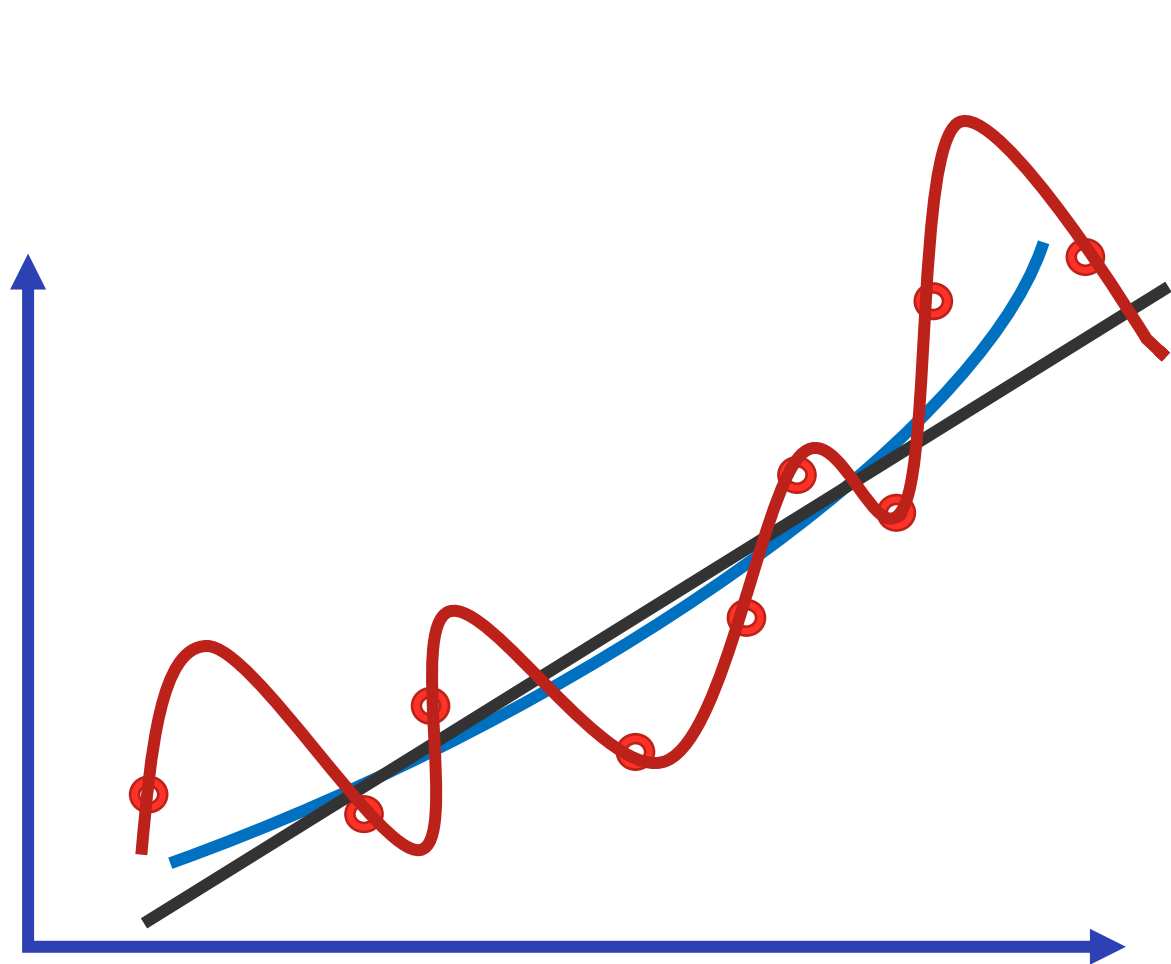
Separar os dados de treino dos dados de teste



Separar os dados de treino dos dados de teste



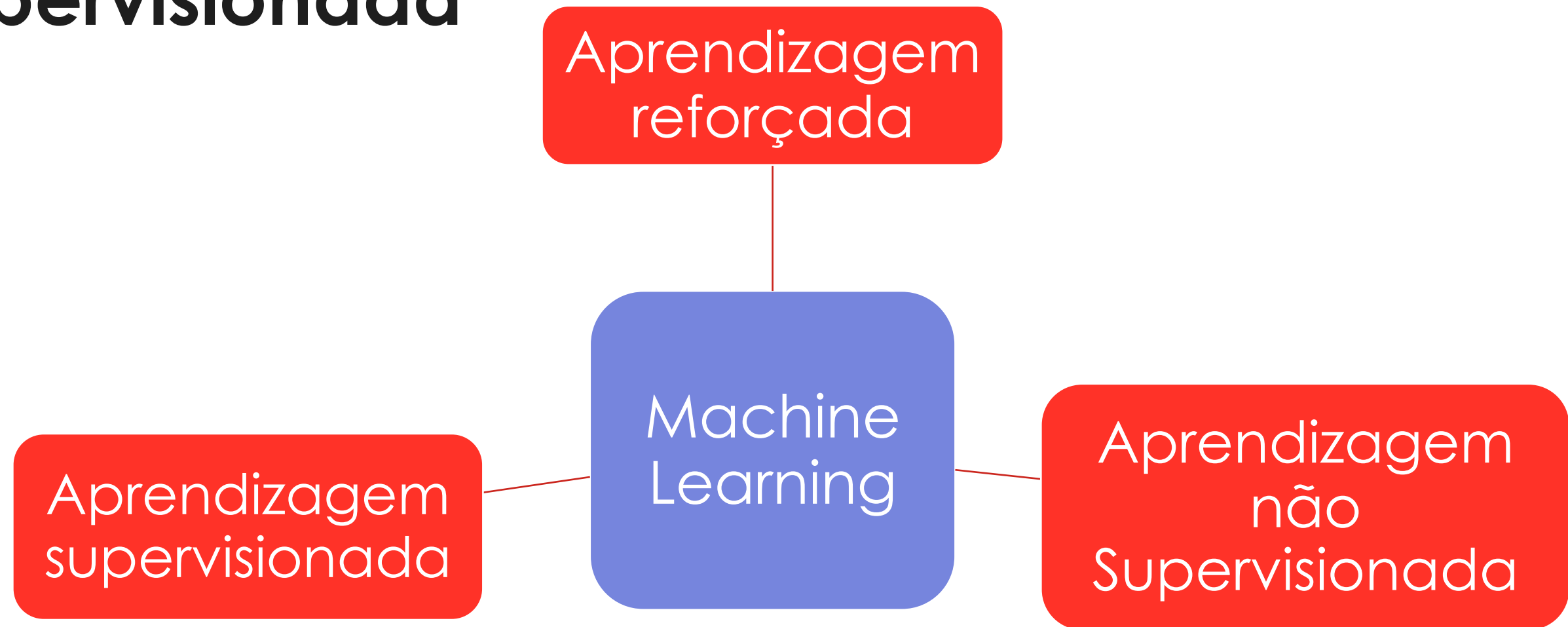
Regressão ou Classificação



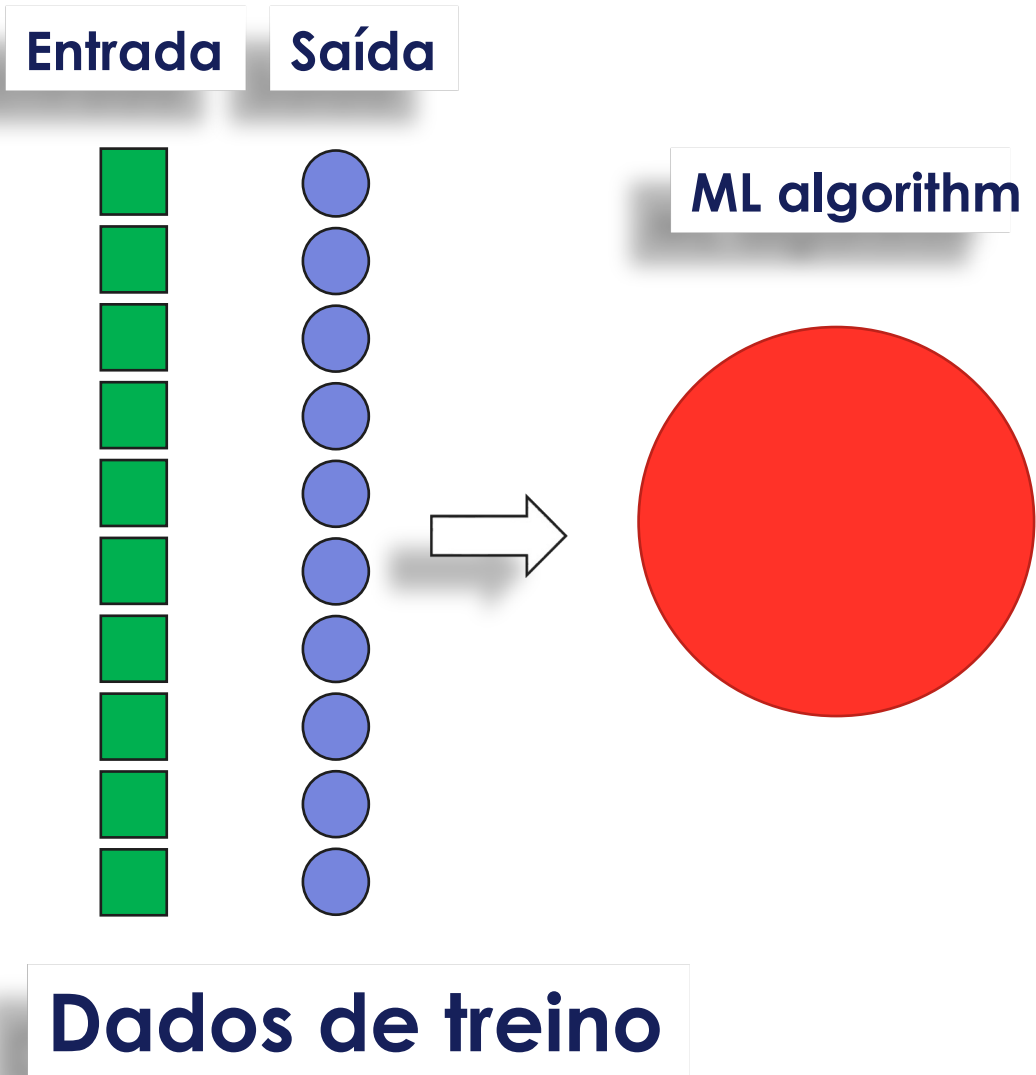
Como verificar a qualidade da classificação?

TAXA DE ERRO = NÚMERO DE ENGANOS[(REAL(X) ≠ MODELO(x))]/TOTAL DE TESTES

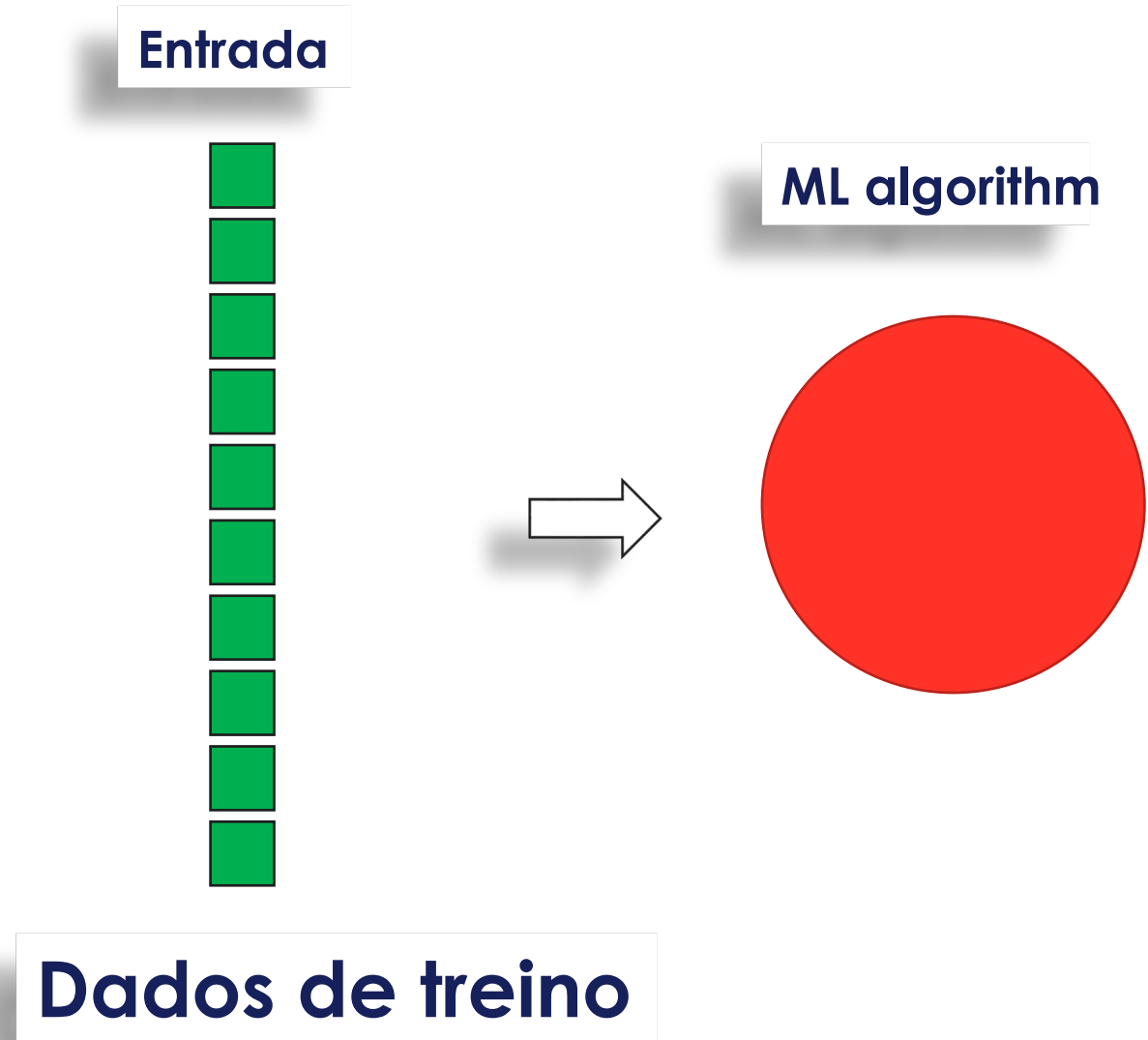
Aprendizagem supervisionada e não supervisionada



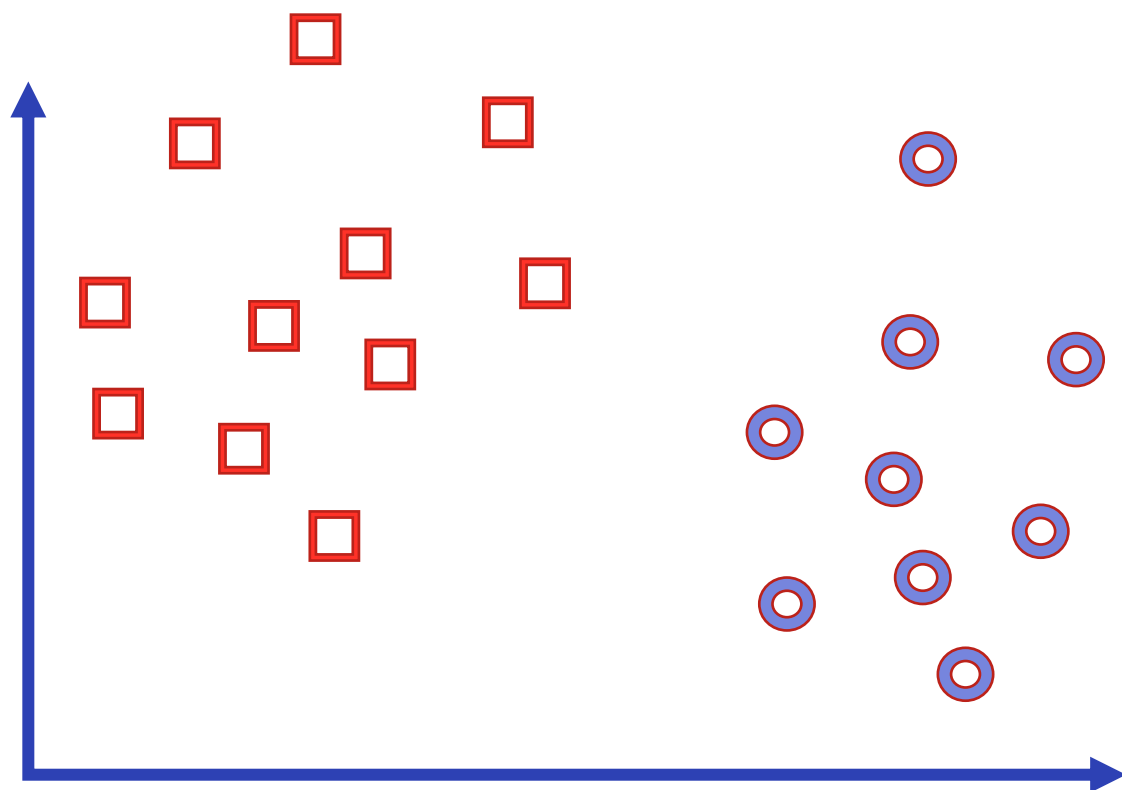
Supervisionada



Não Supervisionada

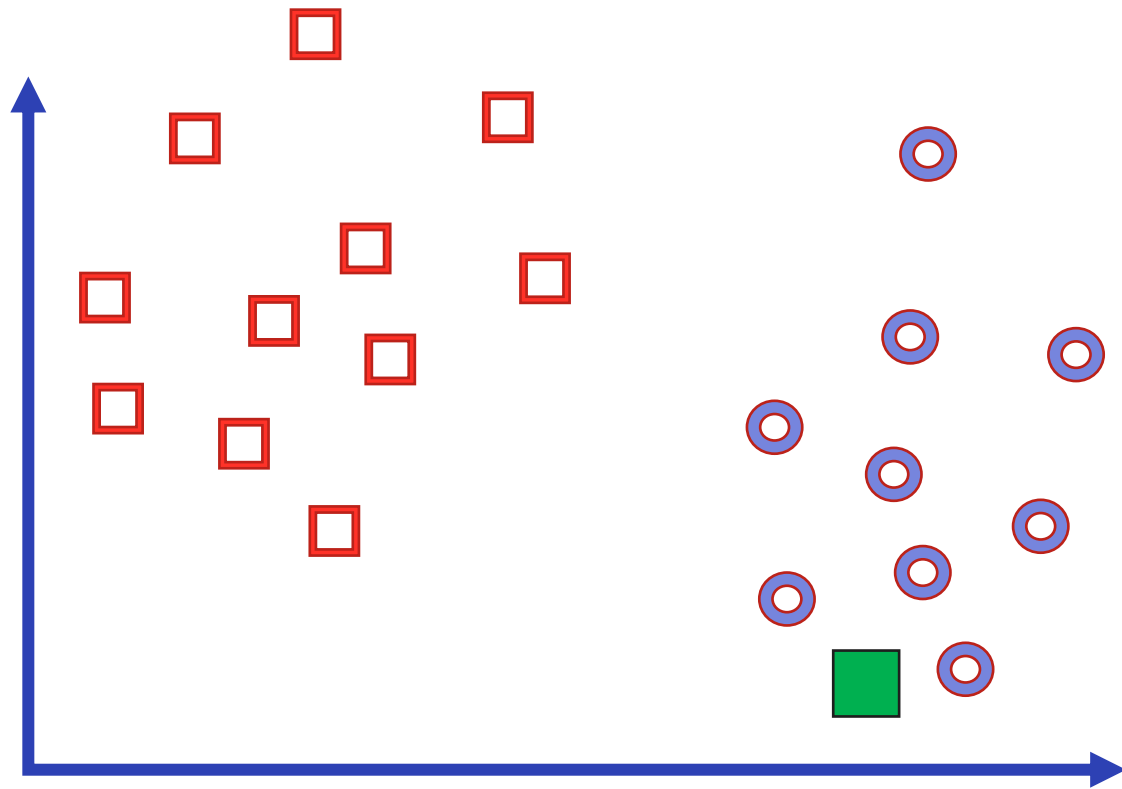


Supervisionada



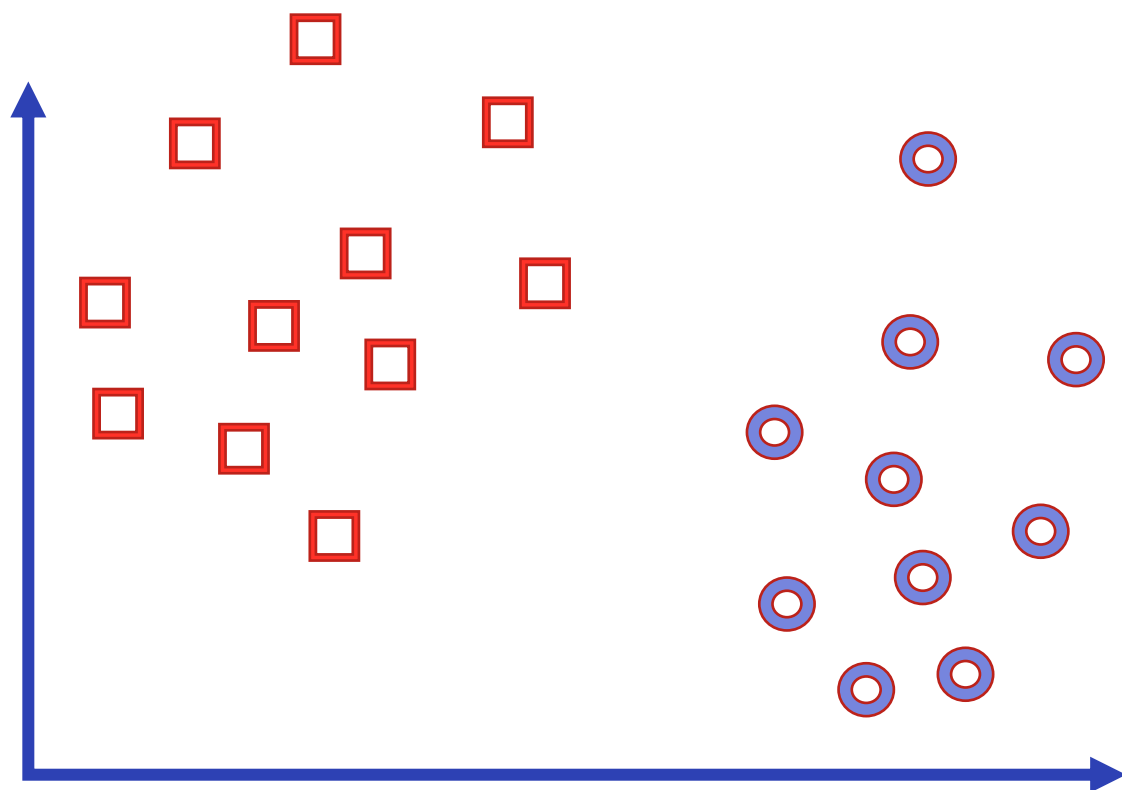
Não Supervisionada

Supervisionada



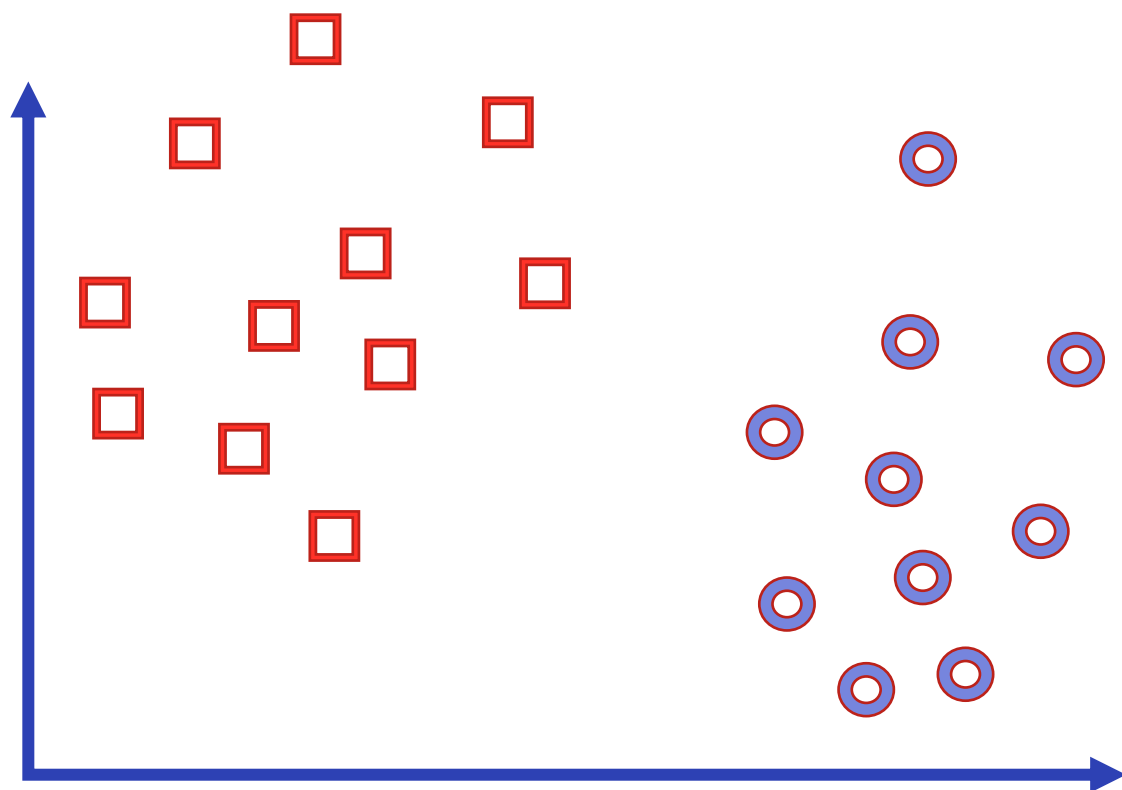
Não Supervisionada

Supervisionada

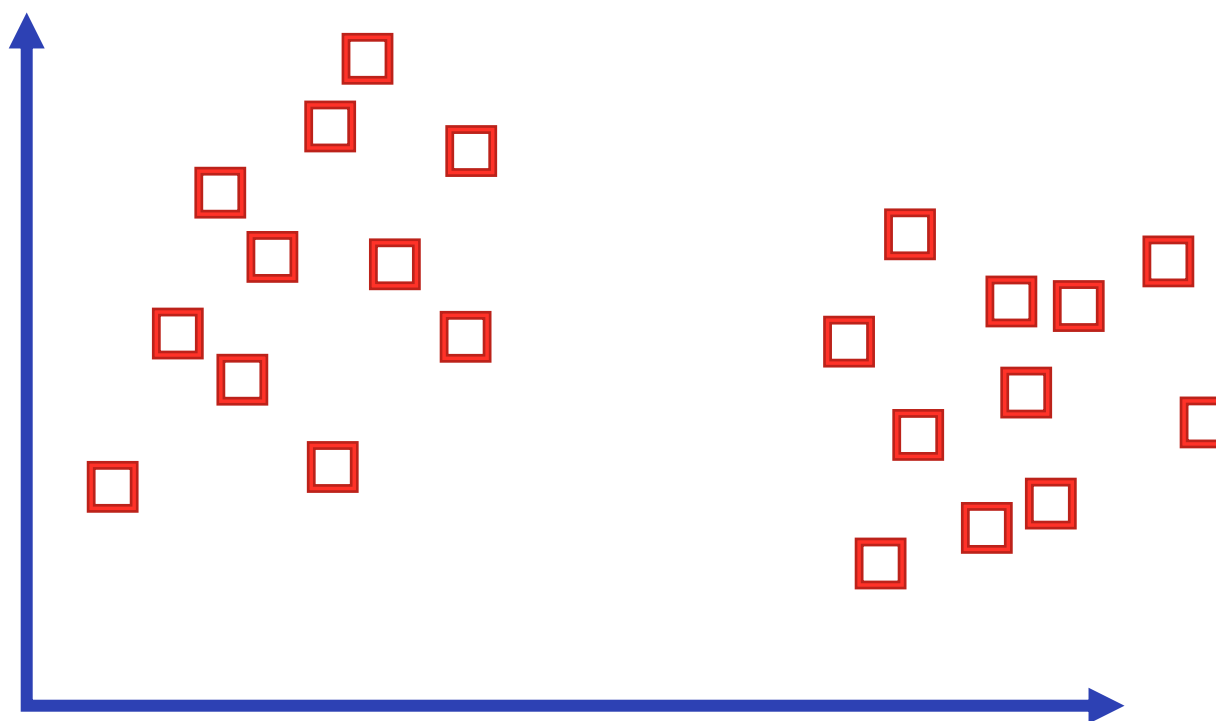


Não Supervisionada

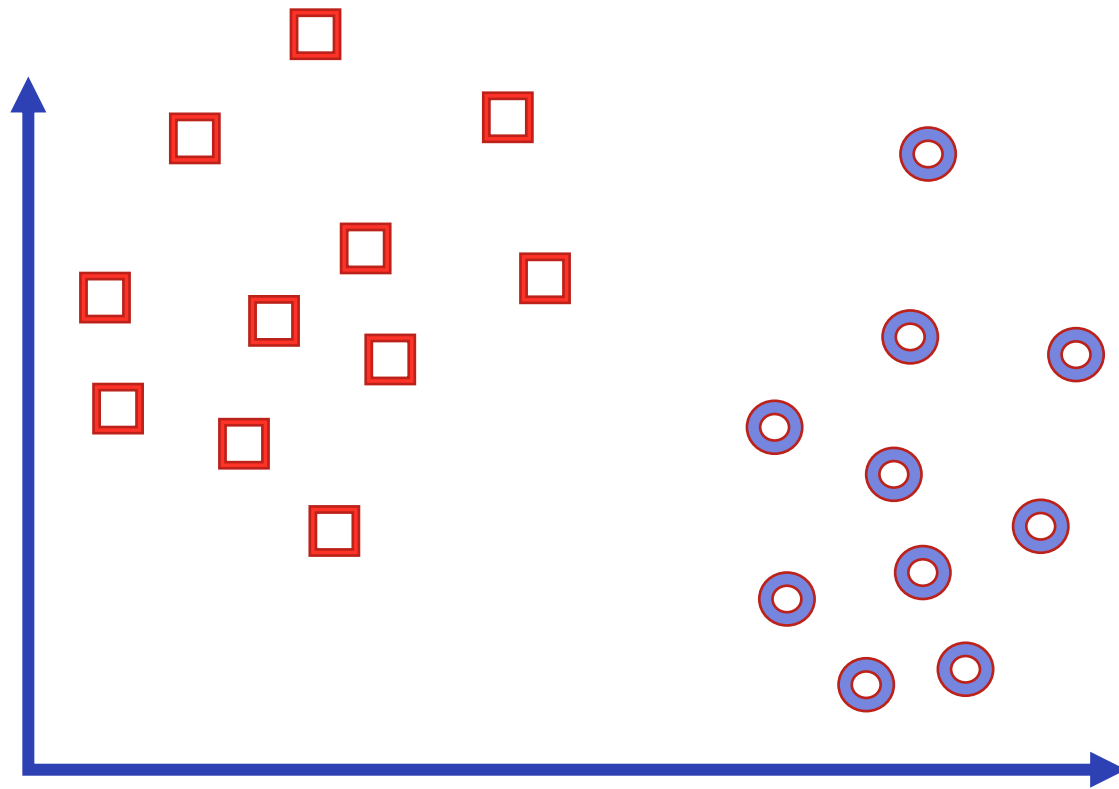
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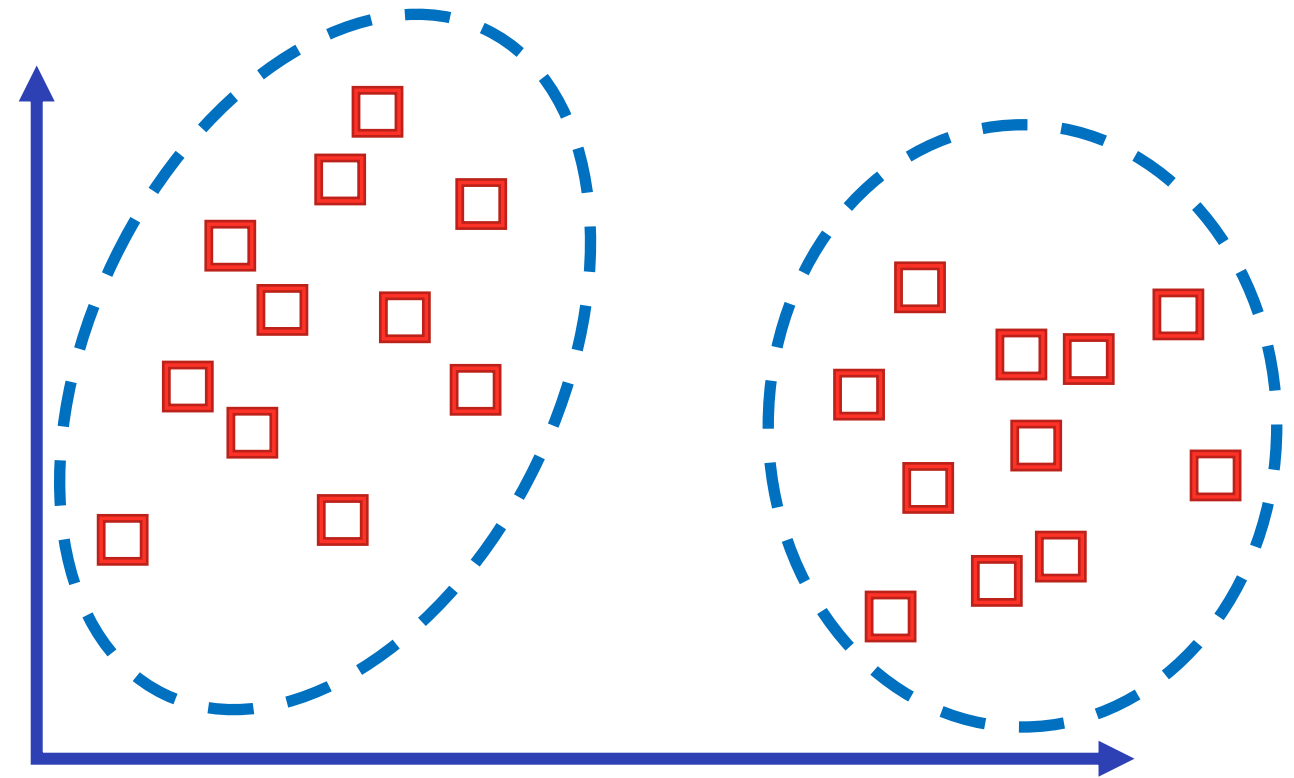
Não Supervisionada



Supervisionada



Não Supervisionada



Supervisionada

- Reconhecimento de escrita
- Detecção de spam
- Processamento natural de texto
- Reconhecimento de objetos

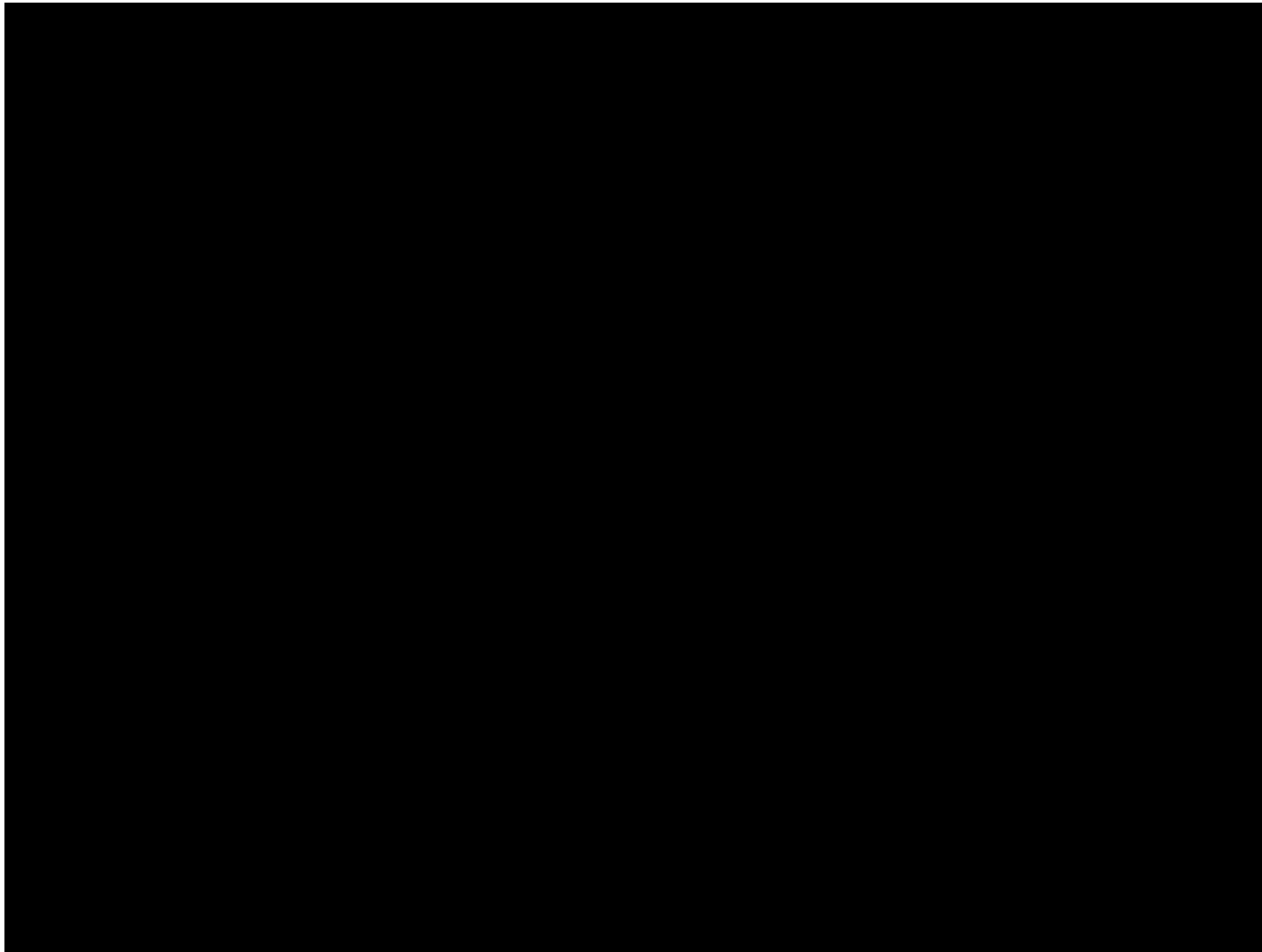
Não Supervisionada

- Agrupamento de dados
- Redução de dimensionalidade
- Detecção de anomalias

Aprendizagem reforçada



Aprendizagem reforçada

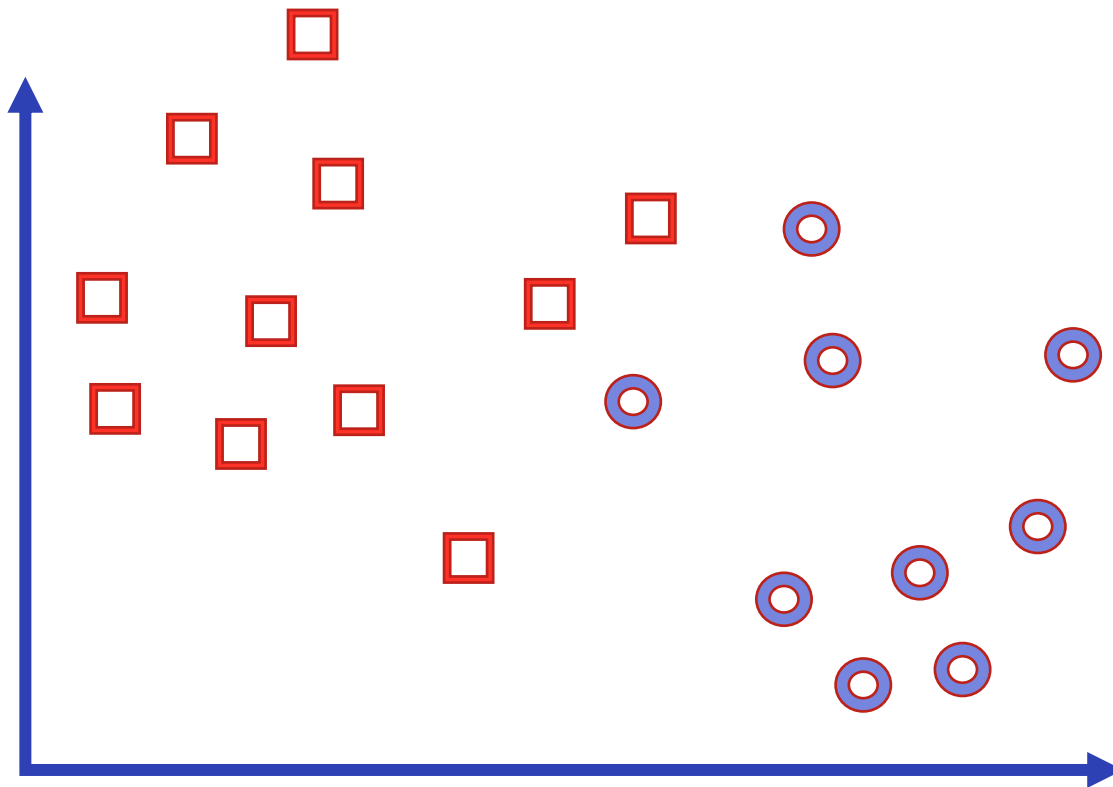


V. Mnih et al. Nature 518, 529 (2015)

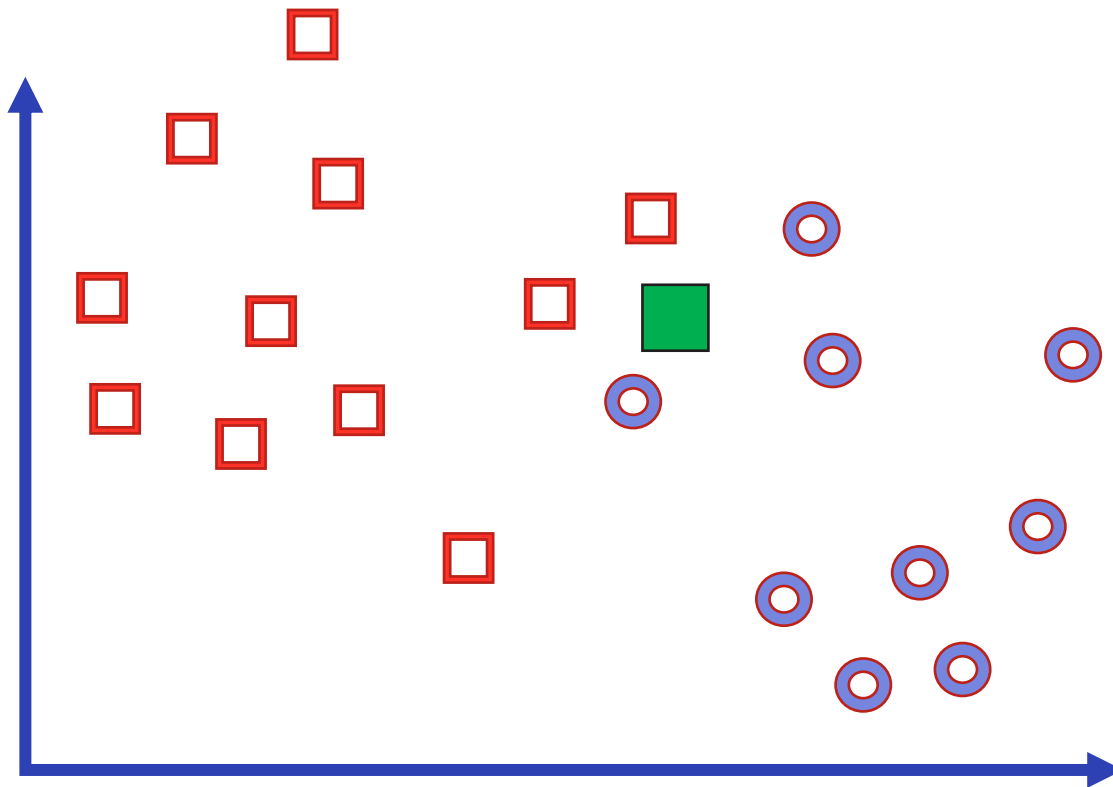
Aprendizagem reforçada



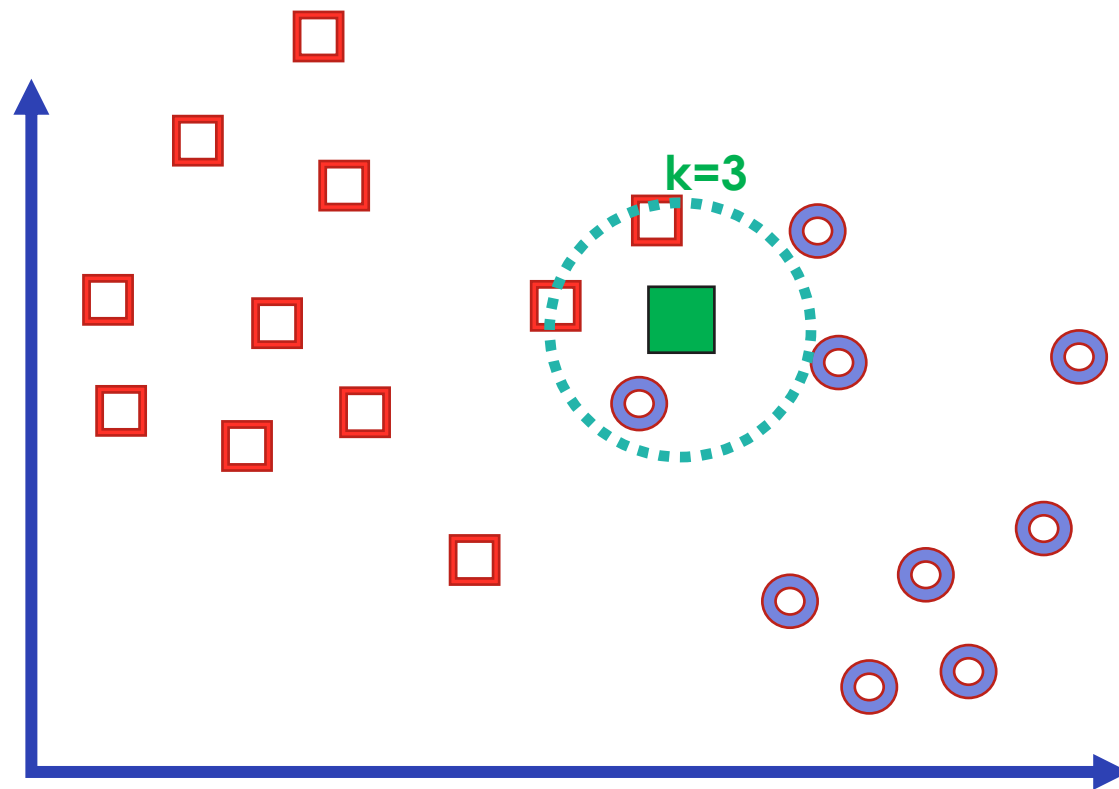
Aprendizagem supervisionada: Algoritmo de k-nearest neighbors



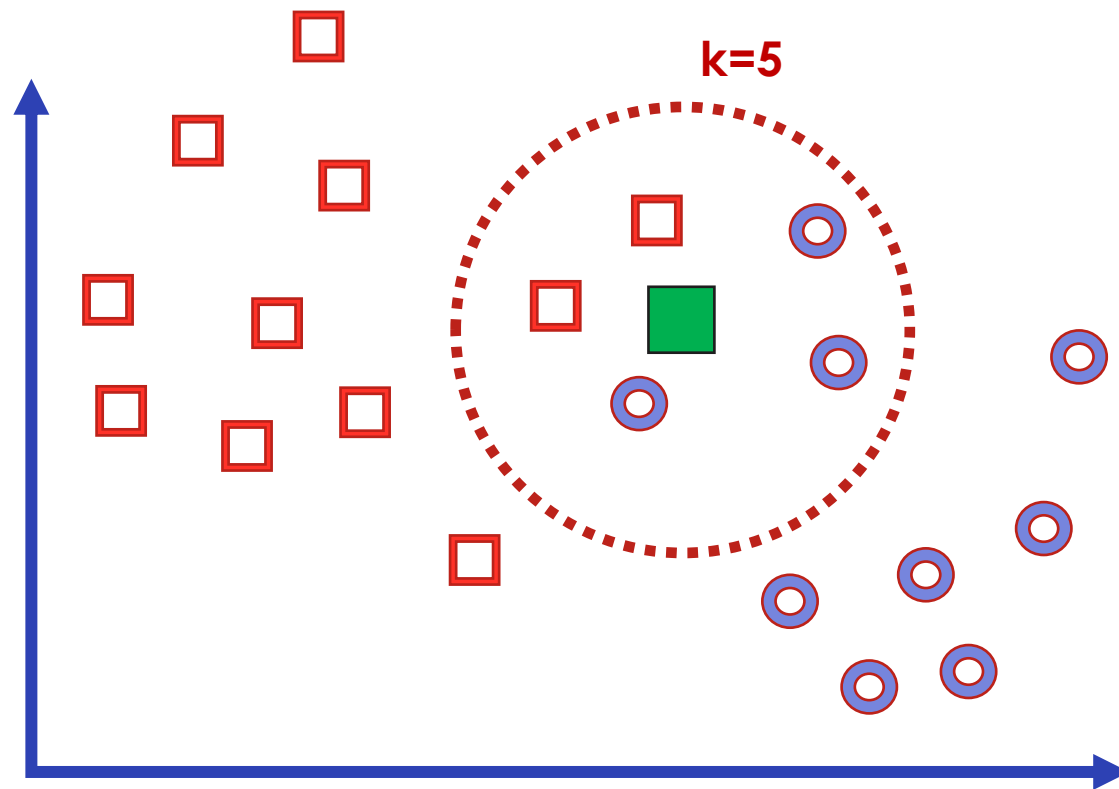
Aprendizagem supervisionada: Algoritmo de k-nearest neighbors



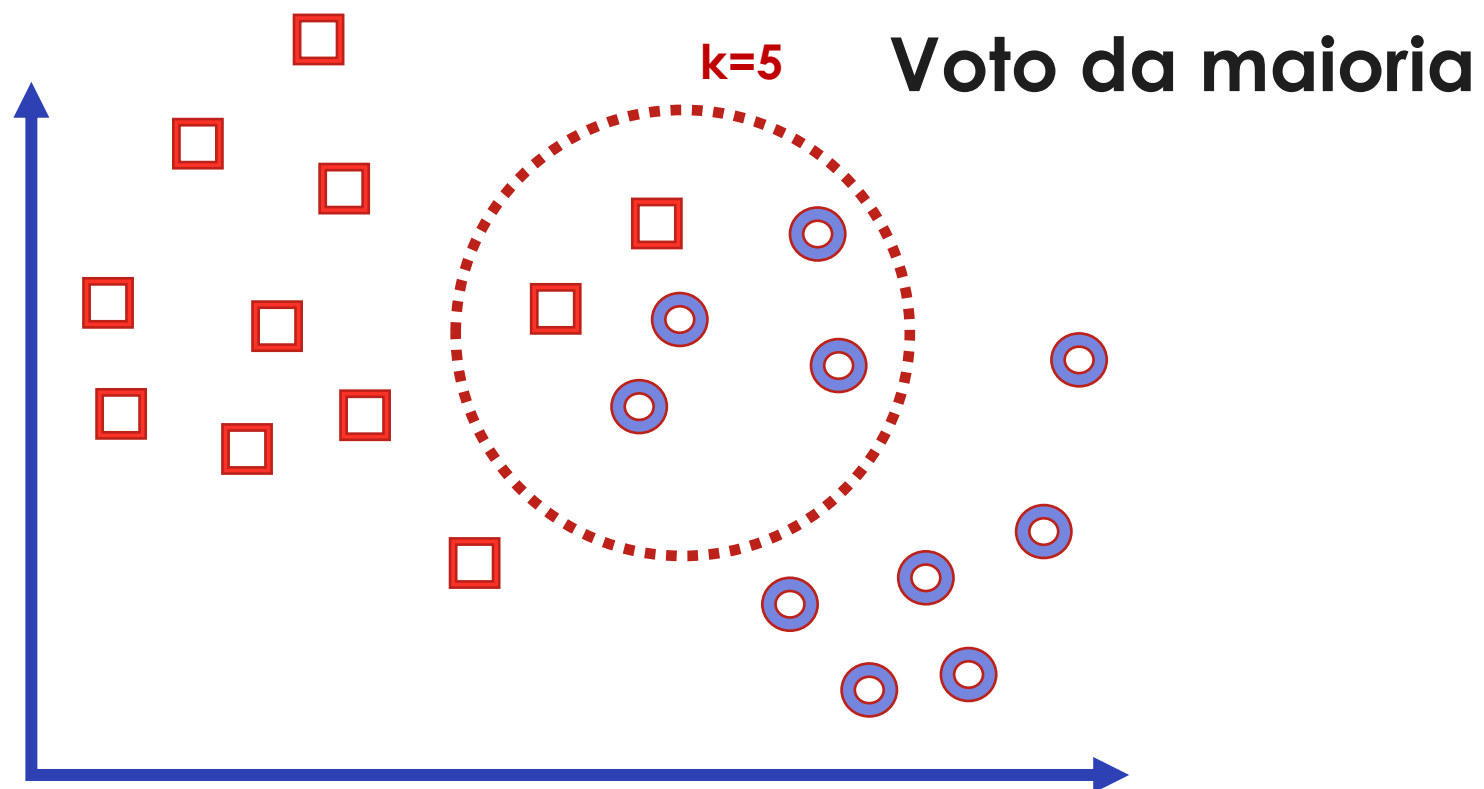
Aprendizagem supervisionada: Algoritmo de k-nearest neighbors



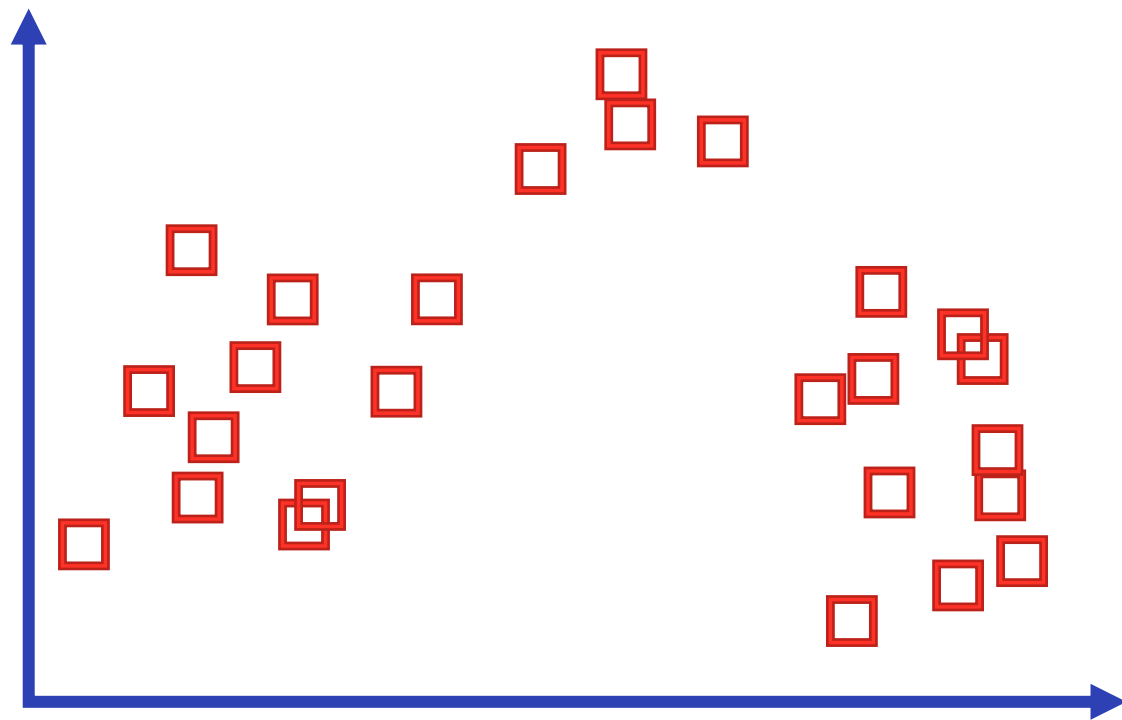
Aprendizagem supervisionada: Algoritmo de k-nearest neighbors



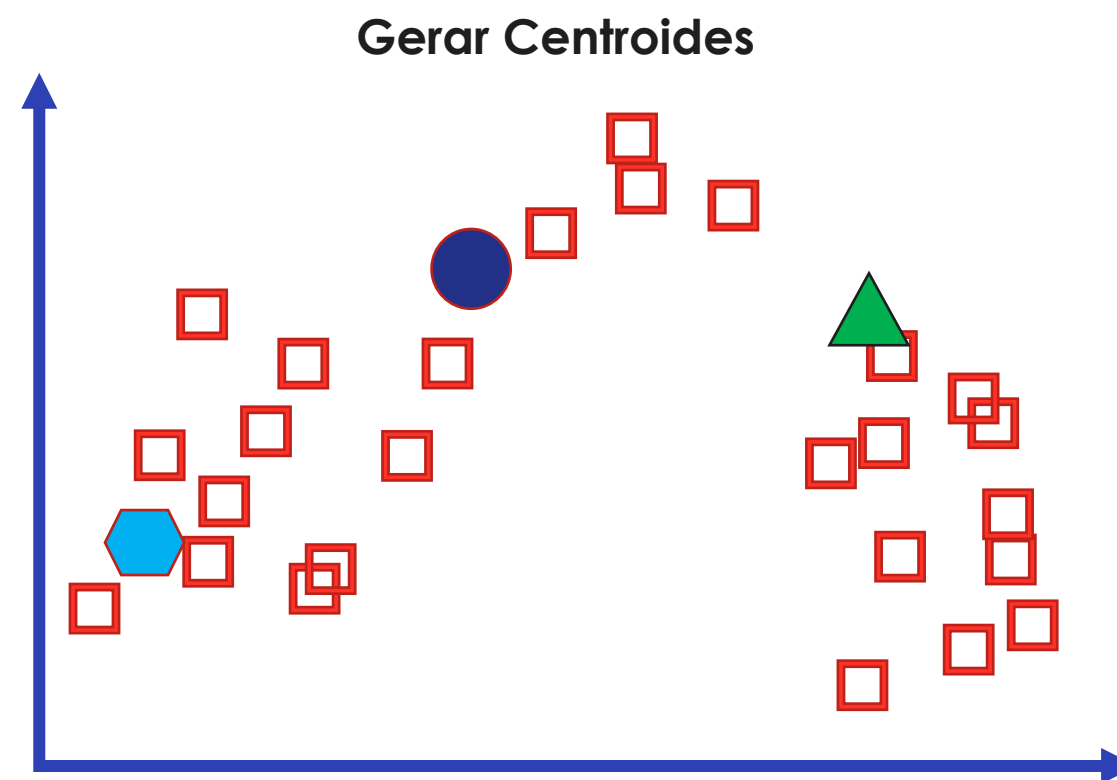
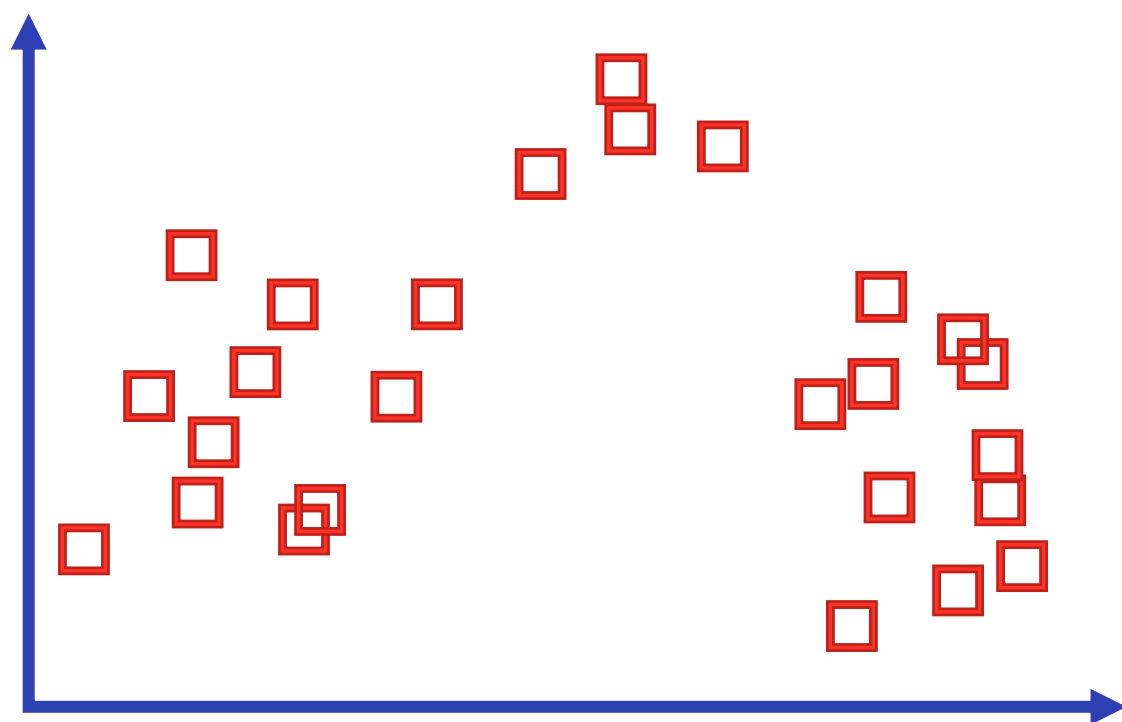
Aprendizagem supervisionada: Algoritmo de k-nearest neighbors



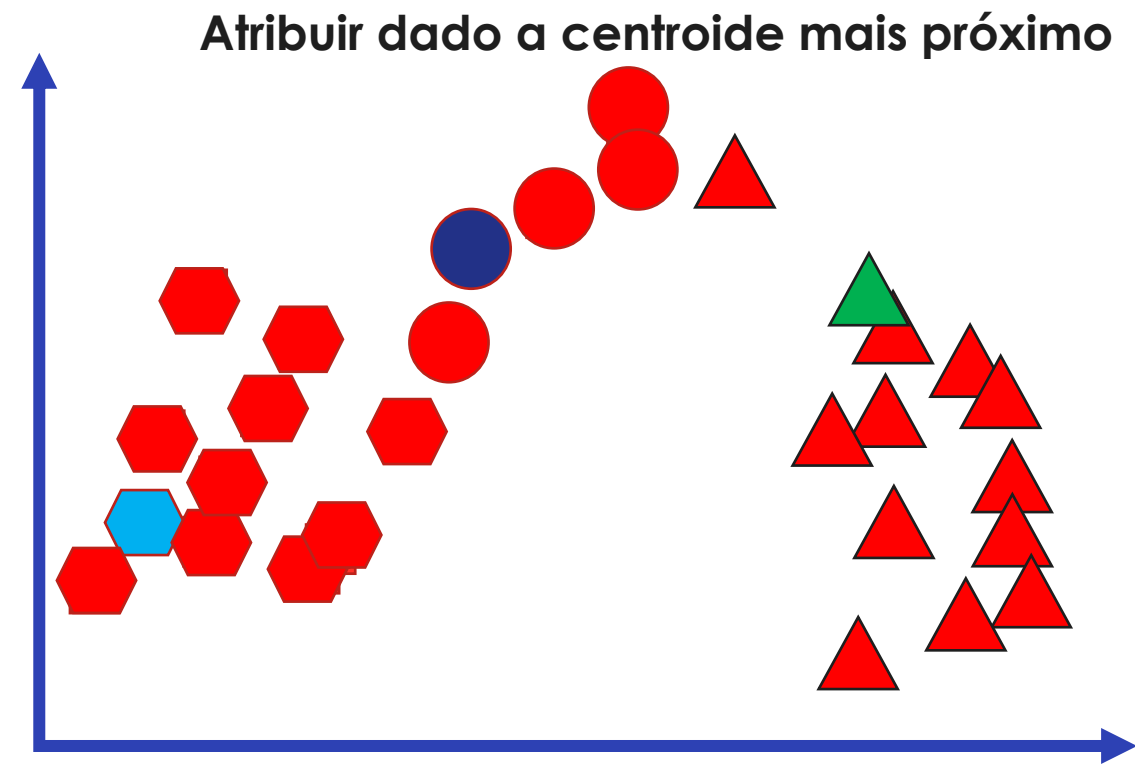
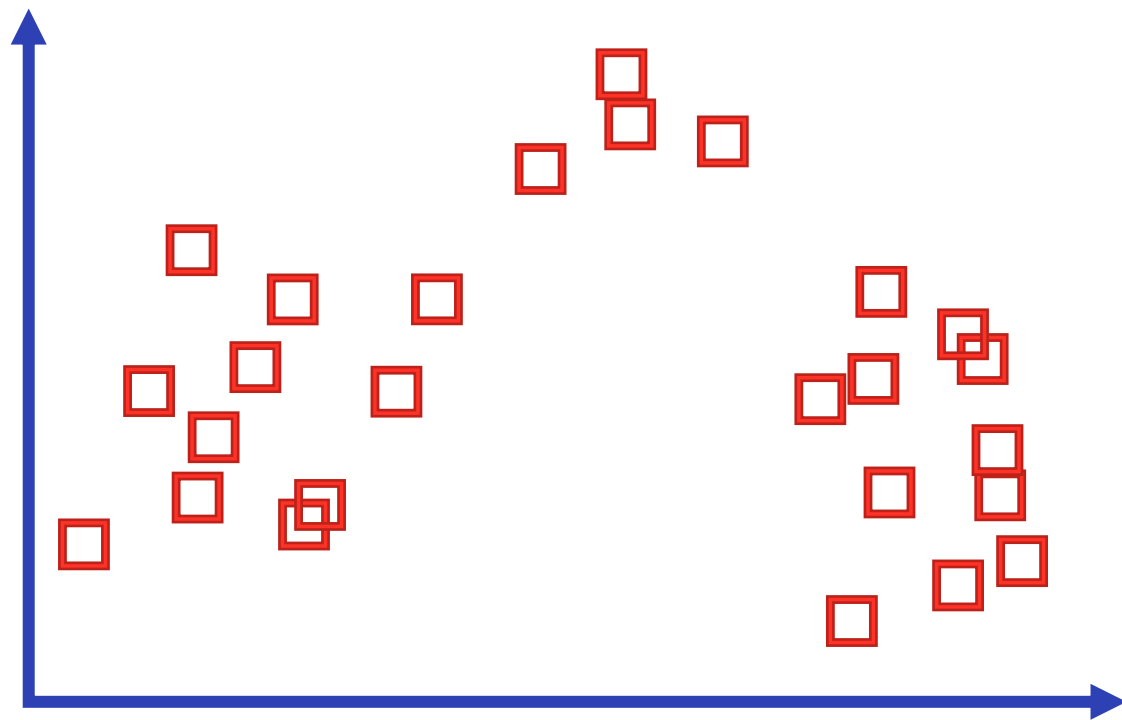
Aprendizagem não supervisionada: Algoritmo de k-means



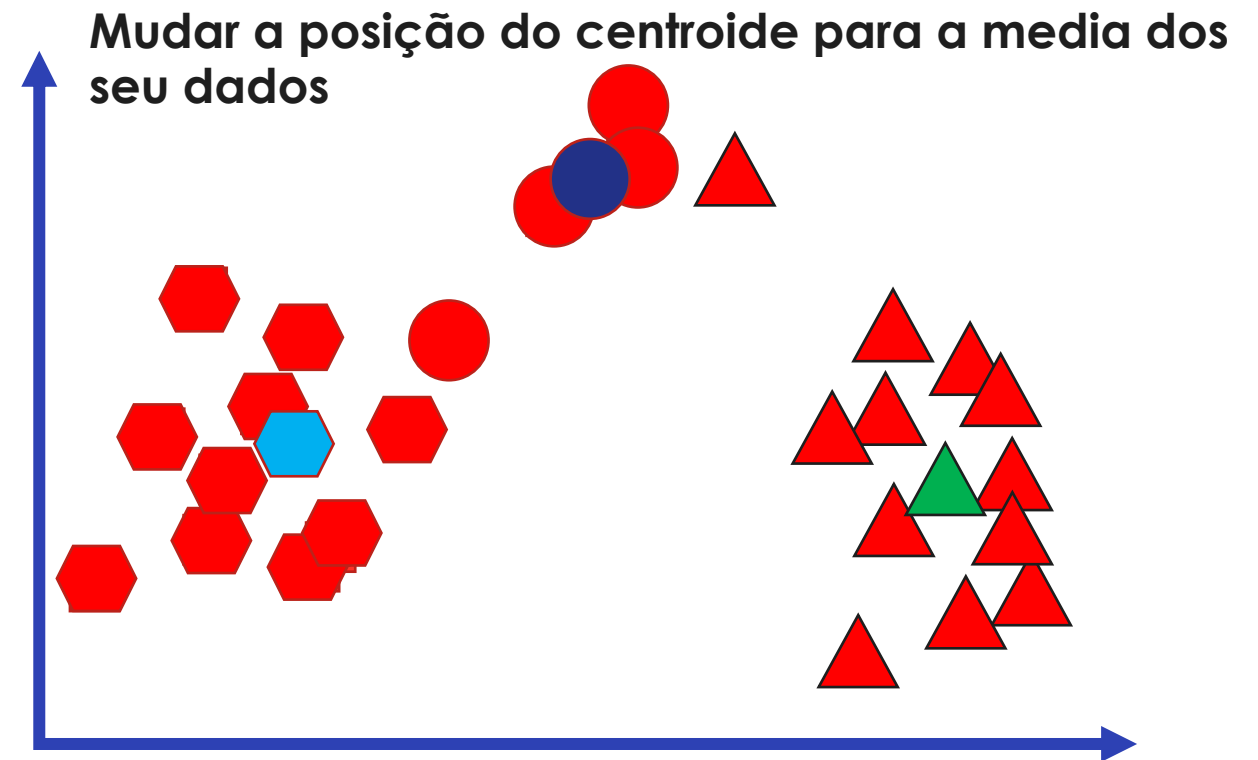
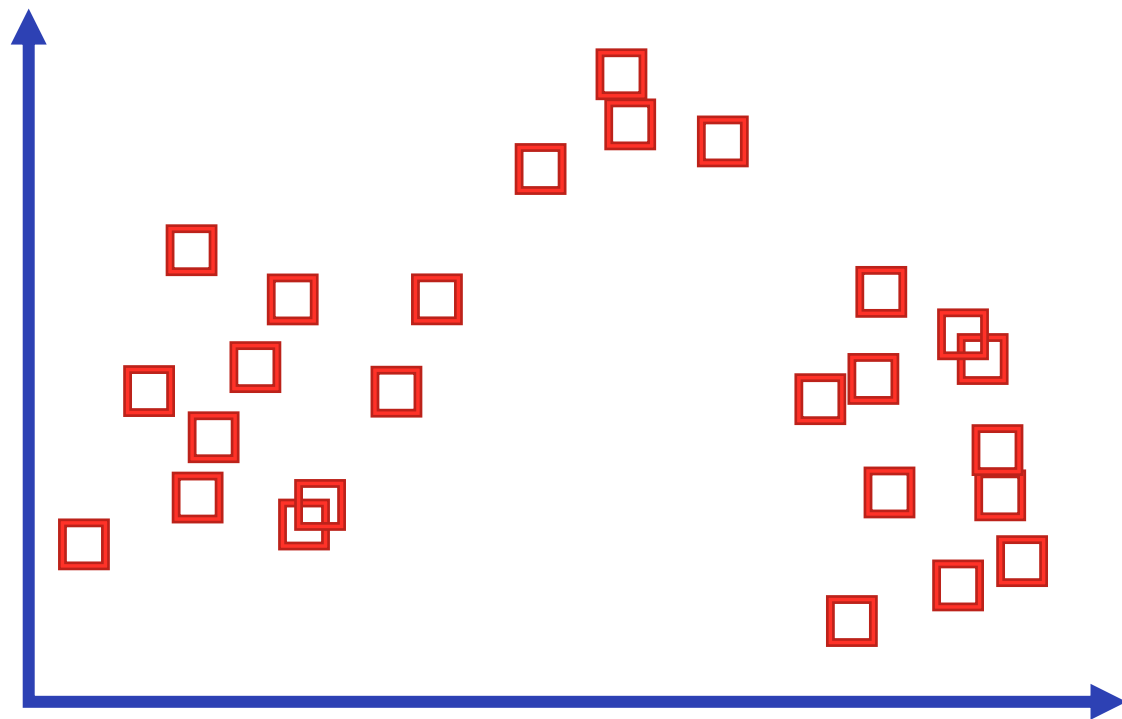
Aprendizagem não supervisionada: Algoritmo de k-means



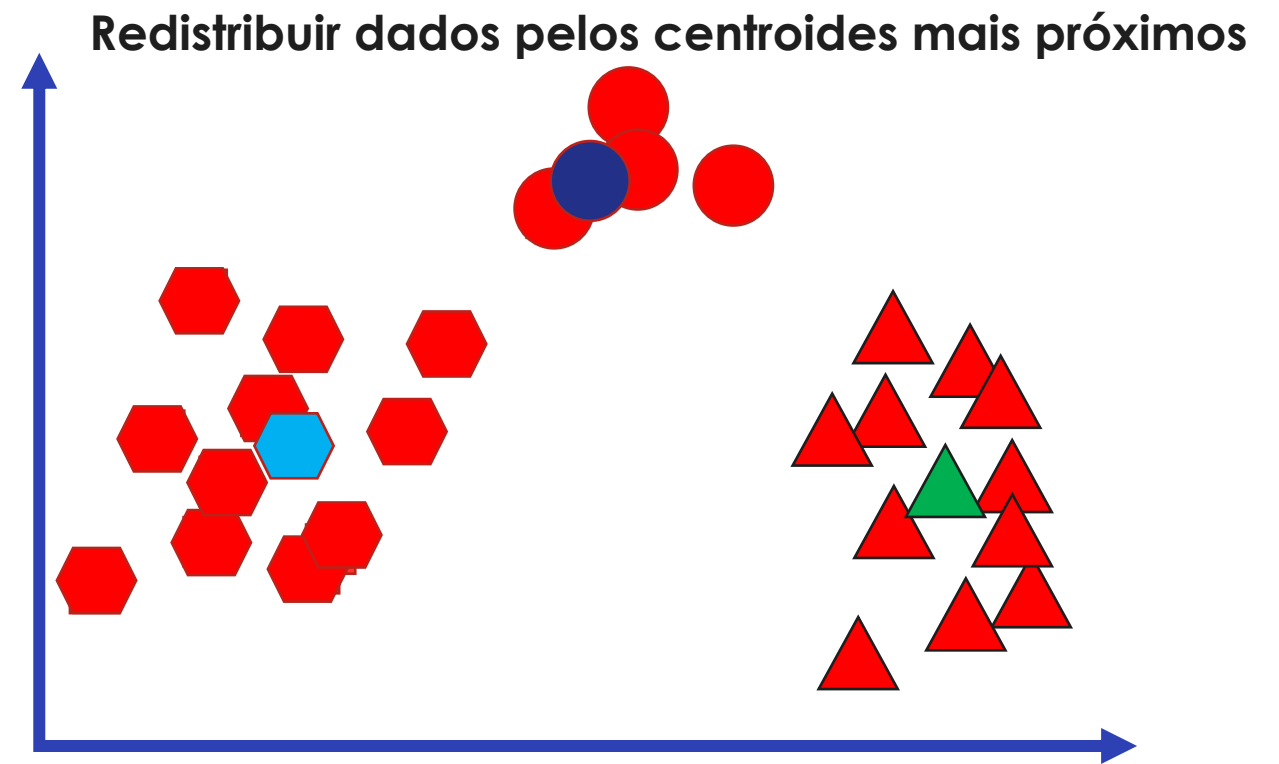
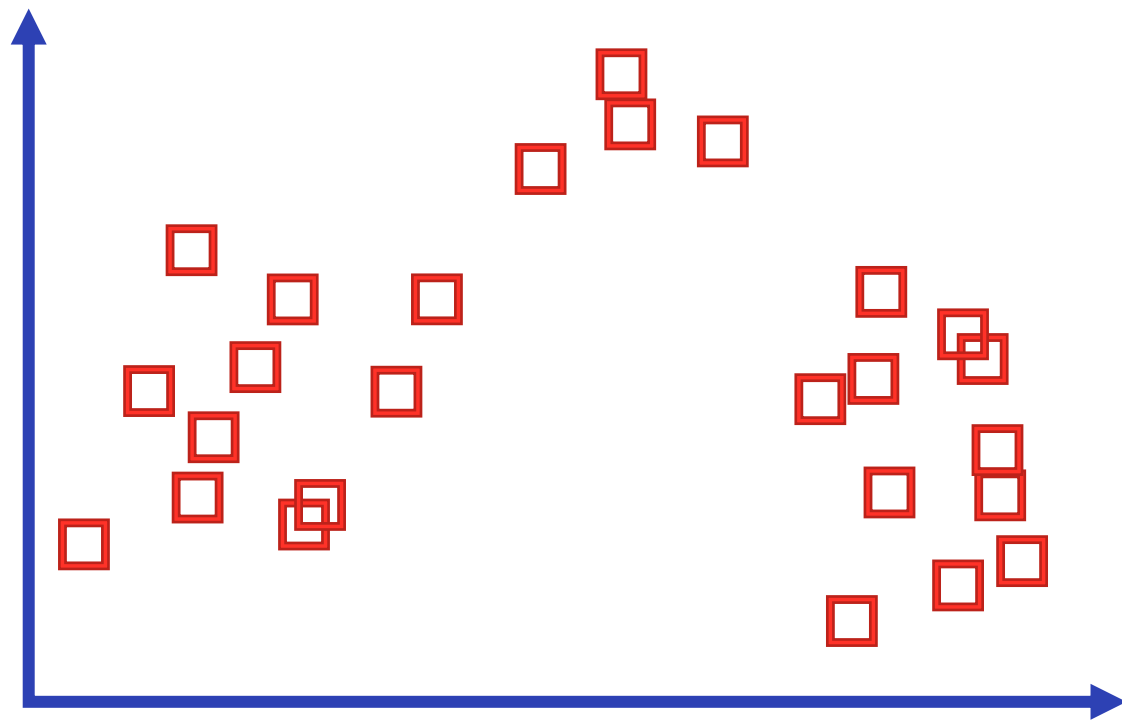
Aprendizagem não supervisionada: Algoritmo de k-means



Aprendizagem não supervisionada: Algoritmo de k-means



Aprendizagem não supervisionada: Algoritmo de k-means



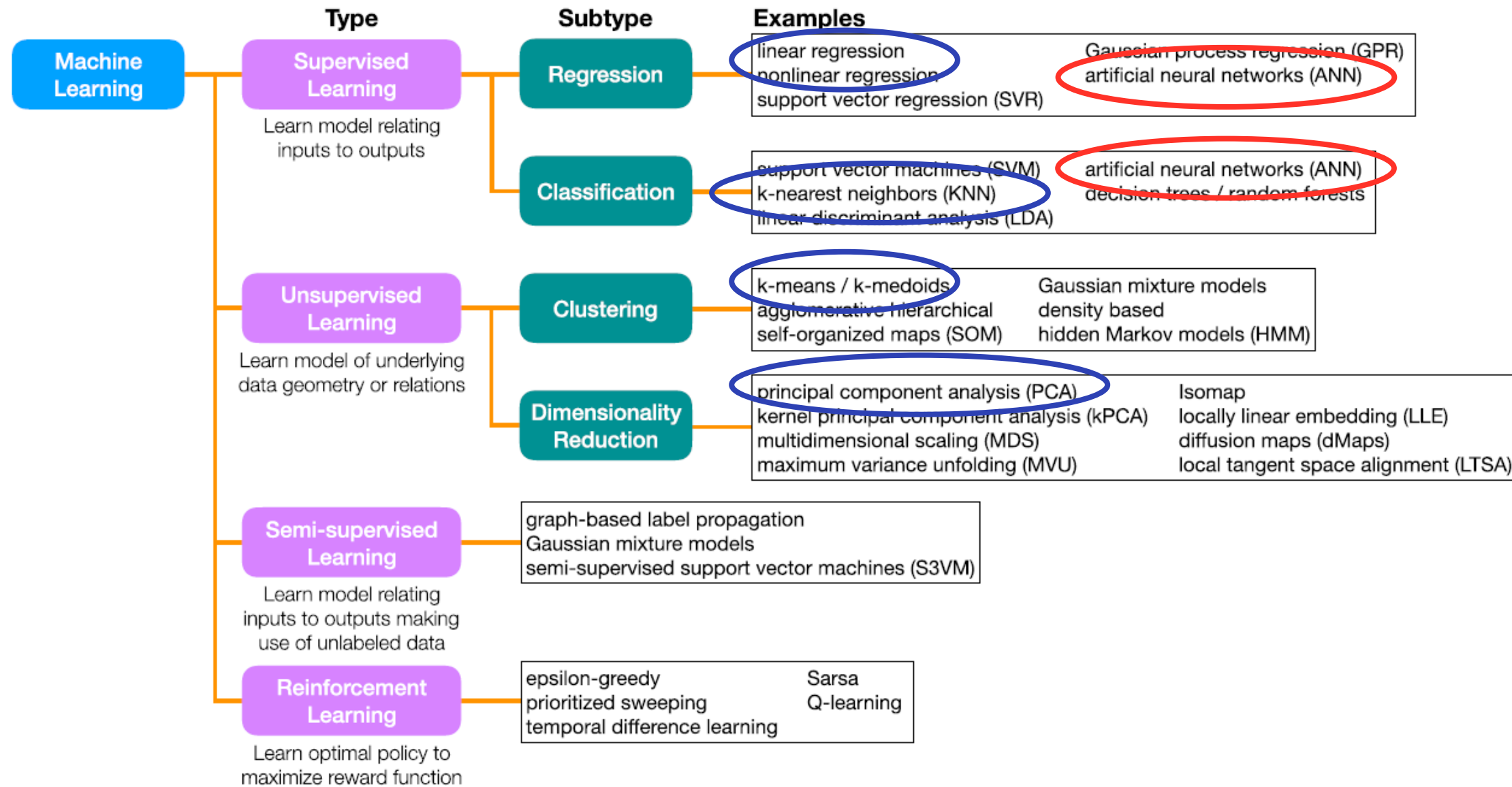


Figure 1. Taxonomy of machine learning algorithms.